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CHAPTER ONE : INTRODUCTIONS

1.0 Introduction

The global manufacturing sector is increasingly focusing on sustainable production, efficient resource utilization, waste reduction, and environmental protection. One of the major contributors to industrial waste generation is the plastics manufacturing industry, particularly the polymer processing sector. Injection moulding industries generate significant amounts of plastic waste during machine startup, process adjustments, and defective product rejection. Polyethylene Terephthalate (PET), which is widely used in beverage packaging and container production, contributes considerably to industrial plastic waste generation due to rejected PET preforms produced during manufacturing operations.

Globally, plastic production exceeded 430 million tonnes in 2023, with packaging applications accounting for a major share of plastic usage (UNEP, 2023). However, only a small percentage of plastic waste is successfully recycled, resulting in increasing environmental pollution and resource depletion. Within manufacturing industries, many PET preform rejects still retain acceptable material properties but are discarded instead of being recycled and reused.

In Zimbabwe, manufacturing industries continue to face challenges such as high raw material costs, foreign currency shortages, unreliable energy supply, and inefficient waste management systems. According to the Confederation of Zimbabwe Industries (CZI) Manufacturing Sector Survey (2024), manufacturing sector capacity utilization remains below 60%, mainly due to operational inefficiencies and material shortages. Since most polymer raw materials are imported, production costs continue to increase, creating a need for more sustainable and cost-effective manufacturing approaches.

As a result, industries are increasingly adopting circular economy principles that promote recycling, waste minimization, and improved material utilization. Recycling and re-purposing PET preform rejects therefore presents an opportunity to reduce production costs, minimize waste generation, and improve manufacturing sustainability.

This research focuses on the development and evaluation of a recycling and re-purposing strategy for PET preform rejects generated during injection moulding operations. The study combines laboratory experimentation and simulation-based engineering analysis to evaluate the feasibility of recycling PET rejects back into the production system.

1.1 Background to the Study

The global manufacturing industry has expanded rapidly due to industrialization, urbanization, and rising consumer demand. Among the major contributors to this growth is the plastics manufacturing sector, which supports industries such as packaging, automotive, electronics, healthcare, and construction. Plastics are widely used because they are lightweight, durable, corrosion resistant, and relatively inexpensive to produce. According to Plastics Europe (2023), global plastic production increased from about 2 million tonnes in 1950 to over 430 million tonnes annually by 2023. Packaging applications account for nearly 40% of total plastic demand, with Polyethylene Terephthalate (PET) being one of the most commonly used thermoplastics due to its strength, transparency, and recyclability. PET is extensively used in beverage bottle production through injection and blow moulding processes.

Despite these advantages, plastic waste generation has become a major environmental and industrial challenge. The Organisation for Economic Co-operation and Development (2022) reported that global plastic waste exceeds 353 million tonnes annually, yet only about 9% is recycled, while large quantities are land-filled or mismanaged. In addition, the United Nations Environment Programme (2023) estimated that around 11 million tonnes of plastic waste enter aquatic ecosystems each year. These concerns have increased pressure on industries to adopt sustainable manufacturing systems and circular economy practices focused on recycling, waste reduction, and efficient resource utilization.

Injection moulding industries contribute significantly to plastic waste generation through startup rejects, defective products, contamination, and dimensional inaccuracies. In PET bottle manufacturing, rejected PET preforms are commonly produced during machine startup and process instability. Although many rejected preforms still retain acceptable properties, they are often discarded instead of recycled. Studies by Ragaert, Delva, and Van Geem (2017) showed that industrial plastic waste has high recycling potential because it is cleaner than post-consumer waste. Similarly, Awaja and Pavel (2005) highlighted that PET can be effectively recycled through crushing, washing, drying, and pelletizing processes.

Recycling PET provides both environmental and economic benefits. Research by Hopewell, Dvorak, and Kosior (2009) found that recycled PET can reduce energy consumption by nearly 79% compared to virgin PET production, while Shen, Worrell, and Patel (2010) reported significant reductions in greenhouse gas emissions. The Ellen MacArthur Foundation (2017) also estimated that plastic packaging worth billions of dollars is lost annually after single-use applications, emphasizing the importance of recycling systems.

In Zimbabwe, the manufacturing sector faces challenges such as high production costs, foreign currency shortages, obsolete infrastructure, and unreliable energy supply. The Confederation of Zimbabwe Industries Manufacturing Sector Survey (2024) reported that manufacturing capacity utilization remains below 60%, while many industries experience difficulties importing raw materials and maintaining stable production due to power shortages. Polymer processing industries are heavily affected because most thermoplastic resins are imported. Furthermore, the Environmental Management Agency (2023) reported increasing plastic waste accumulation caused by inadequate recycling infrastructure and weak waste management systems.

Despite the growing importance of PET recycling, limited research has focused on integrating PET preform recycling into injection moulding operations under Zimbabwean manufacturing conditions. Most previous studies emphasize post-consumer recycling and large-scale systems without combining laboratory experimentation and simulation-based analysis. This study therefore seeks to develop and evaluate a recycling and re-purposing strategy for PET preform rejects generated during injection moulding operations to improve sustainability, reduce waste generation, and enhance manufacturing efficiency.

1.2 Problem Statement

Zimbabwe's plastics manufacturing sector faces increasing challenges of PET waste generation, high raw material costs, dependence on imported materials, and poor recycling systems. Injection moulding industries continuously discard rejected PET preforms despite their recyclable potential, leading to material losses and environmental pollution. In addition, energy shortages and foreign currency constraints continue to reduce manufacturing efficiency and sustainability.

1.3 Aim of the Study

The aim of this research was to explore a recycling and recovery strategy for rejected PET preforms generated during injection moulding operations in order to reduce plastic waste, improve material utilization, and enhance manufacturing sustainability and operational efficiency.

1.4 Research Objectives

- 1.4.1 To identify the major causes and characteristics of PET preform rejects generated during injection moulding operations.
- 1.4.2 To evaluate the mechanical and thermal properties of recycled PET blends produced from rejected PET preforms.
- 1.4.3 To determine suitable recycling and re-purposing applications for recycled PET material.
- 1.4.4 To analyze the effects of recycling on production throughput, waste reduction, material utilization, and operational costs.

1.5 Significance of the Study

1.5.1 Industrial Significance : The study will assist manufacturing industries in reducing material waste, improving resource utilization, and lowering operational costs through recycling of PET preform rejects.

1.5.2 Environmental Significance : The research promotes sustainable manufacturing and circular economy practices by reducing plastic waste disposal and environmental pollution.

1.5.3 Economic Significance : The proposed recycling strategy may reduce dependence on imported virgin PET material, thereby reducing production costs and improving profitability.

1.5.4 Academic and Engineering Significance : The study contributes to engineering knowledge by integrating laboratory experimentation with simulation-based manufacturing analysis.

1.5.5 National Significance : The study aligns with Zimbabwe's industrial sustainability and waste management objectives.

1.6 Assumptions of the Study

- 1.6.1 PET rejects are sufficiently clean for mechanical recycling.
- 1.6.2 Laboratory results accurately represent production conditions.
- 1.6.3 Simulation models accurately represent manufacturing system behavior.
- 1.6.4 Production parameters remain relatively stable during experimentation.
- 1.6.5 Recycled PET can be blended with virgin PET without causing major process instability.

1.7 Limitations of the Study

- 1.7.1 Limited industrial production data availability
- 1.7.2 Limited access to advanced testing equipment
- 1.7.3 Financial limitations affecting pilot-scale implementation
- 1.7.4 Time constraints for long-term testing
- 1.7.5 Variability in reject generation rates

1.8 Delimitations of the Study

This research, conducted at Willowton Group Zimbabwe(WGZ), focused on improving manufacturing sustainability through recycling of rejected PET preforms generated during injection moulding operations. It included:

1.7.1 PET Preform Rejects: Evaluation of rejected PET preforms generated during injection moulding processes.

1.7.2 Mechanical Recycling: Assessment of crushing, shredding, and recycling processes for PET waste recovery.

1.7.3 Laboratory Experimentation: Testing and analysis of recycled PET material properties under laboratory conditions.

1.7.4 Simulation-Based Analysis: Evaluation of manufacturing performance indicators such as waste reduction, material utilization, and operational efficiency using simulation tools.

1.9 Chapter Summary

This chapter introduced the research study on the recycling and re-purposing of PET preform rejects generated during injection moulding operations. The chapter presented the global and local background of plastic manufacturing and waste management, highlighting the increasing need for sustainable manufacturing systems.

The chapter further identified the research problem, established the aim and objectives of the study, and discussed the significance, scope, assumptions, limitations, and delimitations of the research. Additionally, the chapter highlighted the integration of laboratory experimentation and simulation-based analysis as the core engineering approach for evaluating PET recycling feasibility.

The next chapter presents a detailed review of literature related to PET recycling technologies, injection moulding waste management, circular economy principles, manufacturing simulation systems, and sustainable manufacturing approaches.

CHAPTER 2: LITERATURE REVIEW

2.0 Introduction

This chapter reviews literature related to the recycling and re-purposing of rejected Polyethylene Terephthalate (PET) preforms generated during injection moulding operations. The review focuses on global plastic manufacturing and waste generation, PET material characteristics, injection moulding defects, PET recycling technologies, sustainable manufacturing, circular economy principles, recycled PET applications, material simulation software, and simulation-based manufacturing analysis.

The chapter further examines previous research studies on PET preform defects, recycled PET material performance, and the integration of recycled materials into manufacturing systems. The literature review provides the theoretical and empirical foundation for evaluating the feasibility of recycling PET preform rejects to improve manufacturing sustainability, reduce waste generation, and enhance operational efficiency.

2.1 Global Plastic Manufacturing and Waste Generation

The global plastics manufacturing industry has expanded significantly due to industrialization, urbanization, technological advancement, and increasing consumer demand. Plastics are widely used in packaging, automotive, electronics, healthcare, agriculture, and construction industries because of their lightweight nature, corrosion resistance, durability, flexibility, and relatively low production costs.

According to Plastics Europe (2023), global plastic production increased from approximately 2 million tonnes in 1950 to over 430 million tonnes annually by 2023. Packaging applications account for nearly 40% of total plastic demand, making packaging one of the largest consumers of thermoplastic materials.

Polyethylene Terephthalate (PET) is among the most widely used thermoplastics in packaging applications because of its transparency, strength, dimensional stability, and recyclability. PET is extensively used in beverage bottles, food packaging containers, and pharmaceutical packaging through injection moulding and blow moulding processes.

Despite the advantages associated with plastics, plastic waste generation has become a major environmental and industrial challenge. The Organisation for Economic Co-operation and Development (OECD, 2022) reported that global plastic waste generation exceeds 353 million tonnes annually, while only about 9% is effectively recycled. Large quantities of plastic waste are landfilled, incinerated, or released into aquatic and terrestrial ecosystems.



2.2 Polyethylene Terephthalate (PET) Material

Polyethylene Terephthalate (PET) is a thermoplastic polyester polymer widely used in packaging industries due to its excellent mechanical and thermal properties. PET possesses high tensile strength, impact resistance, transparency, lightweight characteristics, and good gas barrier properties.

PET is commonly processed using injection moulding and blow moulding technologies for the production of beverage bottles and packaging containers. During PET preform manufacturing, molten PET resin is injected into mould cavities to produce preforms that are later reheated and blow molded into bottles.

According to Awaja and Pavel (2005), PET is one of the most recyclable thermoplastics because it can be mechanically reprocessed through washing, shredding, drying, crushing, and pelletizing processes. Recycled PET can be reused in packaging materials, polyester fibres, engineering products, and industrial applications.

Vilaplana and Karlsson (2008) explained that PET materials are sensitive to thermal degradation and hydrolytic degradation during repeated processing cycles. Excessive heat, moisture contamination, and prolonged processing may reduce molecular weight, tensile strength, and thermal stability.

Research by Hopewell, Dvorak, and Kosior (2009) found that recycled PET significantly reduces energy consumption compared to virgin PET production while also reducing greenhouse gas emissions and raw material demand.

Because of its recyclability and wide industrial application, PET has become an important material within circular manufacturing systems aimed at reducing waste and improving sustainability.

2.3 Injection Moulding Operations and PET Preform Defects

Injection moulding is one of the most widely used polymer processing methods in modern manufacturing industries. The process involves melting thermoplastic materials and injecting them into mould cavities under pressure to produce specific shapes and components.

PET preforms are manufactured through injection moulding before being converted into bottles using blow moulding processes. Although injection moulding provides high productivity and dimensional accuracy, several operational conditions can result in product defects and material rejection. Research studies identified the following common PET preform defects:

- Short shots
- Flash formation
- Sink marks
- Air bubbles
- Burn marks
- Warpage
- Contamination
- Discoloration
- Dimensional inconsistencies



These defects are often caused by:

- Improper melt temperature
- Excessive injection pressure
- Poor cooling conditions
- Moisture contamination
- Inadequate drying
- Unstable machine settings
- Poor mould design
- Material degradation

Research on polymer processing showed that PET materials are highly sensitive to moisture contamination during processing. Moisture causes hydrolytic degradation, resulting in brittleness, bubbles, reduced molecular weight, and discoloration.

Studies further showed that machine startup operations contribute significantly to PET preform rejection because temperature, pressure, and material flow conditions remain unstable during startup periods.

Ragaert, Delva, and Van Geem (2017) found that industrial thermoplastic waste generated during manufacturing operations possesses high recycling potential because it is cleaner and less contaminated than post-consumer waste. This makes industrial PET rejects suitable for mechanical recycling and reprocessing.

Research also demonstrated that process optimization and simulation software can reduce defect generation by improving control of injection pressure, mould temperature, cooling time, and material flow behavior.

2.4 Pet Recycling Technologies

Plastic recycling refers to the recovery and reprocessing of plastic waste into reusable materials and products. PET recycling technologies are generally classified into mechanical recycling, chemical recycling, and energy recovery.

2.4.1 Mechanical Recycling

Mechanical recycling is the most widely used PET recycling method. The process involves collection, sorting, washing, shredding, crushing, drying, melting, and reprocessing plastic waste into reusable material.

According to Grigore (2017), mechanical recycling offers several advantages including:

- Lower production costs
- Reduced energy consumption
- Reduced environmental pollution
- Reduced dependence on virgin raw materials

Mechanical recycling is particularly suitable for industrial PET waste because manufacturing rejects are generally cleaner than post-consumer waste. The main stages of PET mechanical recycling include:

- Collection and sorting
- Washing and contamination removal
- Shredding and crushing
- Drying
- Melting and extrusion
- Pelletizing or pulverizing

Mechanical recycling is economically attractive because it requires lower investment costs compared to chemical recycling technologies. However, repeated processing cycles may reduce polymer quality due to thermal degradation.



2.4.2 Chemical Recycling

Chemical recycling involves breaking polymer chains into monomers or other chemical compounds through chemical reactions such as glycolysis, methanolysis, and hydrolysis. Chemical recycling can produce high-quality recycled material but requires advanced equipment, high energy consumption, and significant investment costs.

Research studies on chemical recycling demonstrated improved purity and restoration of polymer properties compared to mechanical recycling. However, the complexity and cost of chemical recycling limit its application in many developing countries.



2.5 Challenges of Recycling PET Preforms

Several research studies have investigated the challenges associated with recycling PET preforms, recycled PET material performance, and the integration of recycled PET into manufacturing systems. Although PET possesses high recycling potential, researchers identified several technical and operational challenges that affect the quality and performance of recycled PET materials.

Awaja and Pavel (2005) conducted extensive research on PET recycling technologies and found that recycled PET may fail to maintain acceptable mechanical and thermal properties under certain processing and recycling conditions. Their study showed that improper recycling procedures can negatively affect the usability of recycled PET in manufacturing applications.

Vilaplana and Karlsson (2008) studied recycled polymeric materials and found that repeated thermal processing causes degradation of polymer chains, resulting in reduced molecular weight and weakened mechanical properties. Their research highlighted that recycled PET materials are highly sensitive to moisture contamination, excessive processing temperatures, thermal degradation, material contamination, and repeated melting cycles. The researchers emphasized that improper control of these factors significantly reduces recycled PET quality and processing performance.

Research by Ragaert, Delva, and Van Geem (2017) established that although industrial thermoplastic waste is cleaner and easier to process than post-consumer waste, recycling operations still face challenges related to maintaining consistent material quality and preventing contamination during reprocessing.

Studies on PET bottle recycling systems conducted by Shen, Worrell, and Patel (2010) further identified challenges associated with maintaining material performance while reducing environmental impacts during recycling operations.

Research on injection moulding operations identified several major causes of PET preform defects and recycling difficulties including improper drying, temperature fluctuations, excessive injection pressure, poor cooling conditions, unstable process parameters, and material contamination.

Researchers found that PET materials are highly sensitive to moisture during processing. Inadequate drying causes hydrolytic degradation during melting, leading to brittleness, bubble formation, reduced molecular weight, discoloration, and reduced mechanical strength.

Experimental studies on recycled PET blends also revealed that increasing recycled PET content may negatively affect material properties. Many studies focused on recycled PET blending ratios ranging between 5% and 30%, with findings indicating that lower recycled PET content maintains better transparency and strength while higher recycled PET content may reduce tensile strength and thermal stability. Researchers also found that excessive recycled PET content may increase discoloration and brittleness, while poor temperature control accelerates material degradation. Inconsistent blending was also identified as a major cause of process instability during manufacturing operations.

Although blending recycled PET with virgin PET improves processing stability, researchers noted that achieving uniform mixing and consistent product quality remains a major challenge during industrial-scale production. Some studies also investigated pulverized PET powder blends and found that finely processed recycled PET waste can be reintroduced into manufacturing systems at controlled recycling ratios. However, challenges such as particle size variation, uneven mixing, and thermal degradation still affect manufacturing performance and product quality.

Researchers generally concluded that the major challenges of recycling PET preforms include thermal degradation during repeated processing, moisture sensitivity, reduction in molecular weight, reduced transparency, reduced tensile strength, material contamination, difficulty in maintaining consistent product quality, and process instability at high recycled PET ratios. These challenges demonstrate the importance of proper process optimization, controlled recycling conditions, and experimental evaluation when integrating recycled PET materials into preform manufacturing systems.

2.6 Applications of Recycled PET Material

Research studies have shown that recycled PET (rPET) can be reused in numerous industrial and engineering applications depending on recycling conditions and material quality.

2.6.1 Reuse of Recycled PET Material

One of the most important applications of recycled PET material is its reuse in the production of new PET preforms through blending with virgin PET resin. Research studies have shown that rejected PET preforms generated during injection moulding operations can be mechanically recycled, processed into flakes, pellets, or powder, and reintroduced into manufacturing systems at controlled blending ratios.

This recycling approach supports closed-loop manufacturing systems where waste materials generated during production are recovered and reused instead of being discarded. Researchers identified this approach as an effective strategy for reducing raw material consumption, minimizing manufacturing waste, and improving sustainability in polymer processing industries.

Experimental studies on recycled PET blends showed that recycled PET can be successfully mixed with virgin PET material while maintaining acceptable processing and mechanical properties when proper processing conditions are followed. Common industrial blending ratios investigated in previous research range between:

- 5% recycled PET and 95% virgin PET
- 10% recycled PET and 90% virgin PET
- 15% recycled PET and 85% virgin PET
- 20% recycled PET and 80% virgin PET

Research findings generally indicate that lower recycled PET ratios maintain better transparency, strength, and thermal stability, while higher recycled PET content may increase the risk of discoloration, reduced molecular weight, brittleness, and thermal degradation.

Awaja and Pavel (2005) found that recycled PET can retain acceptable material properties when contamination and moisture content are properly controlled during recycling and reprocessing operations. Their research emphasized that proper drying is essential because PET is highly sensitive to hydrolytic degradation during melting and injection moulding.

Vilaplana and Karlsson (2008) also reported that repeated thermal processing can reduce polymer quality due to chain scission and molecular degradation. However, blending recycled PET with virgin PET material helps improve process stability and mechanical performance while reducing the negative effects associated with repeated recycling cycles.

Research on injection moulding operations further showed that recycled PET blends can be used in preform manufacturing without causing major process instability when processing parameters such as melt temperature, injection pressure, cooling time, and moisture control are properly optimized.

Several studies also investigated the use of pulverized recycled PET powder blended with virgin PET resin. The findings suggested that finely processed recycled PET particles improve mixing uniformity and material flow behavior during injection moulding operations.

Advantages of Reusing Recycled PET Material

Reusing recycled PET in PET preform production reduces raw material costs and minimizes plastic waste by reintroducing rejected preforms into the manufacturing cycle. This supports sustainable and closed-loop manufacturing systems while reducing environmental pollution and dependence on virgin plastics. Blending recycled PET with virgin PET also improves material utilization and can maintain acceptable processing and mechanical properties when proper drying and processing conditions are applied. Finely processed recycled PET powder further improves material mixing and flow behavior during injection moulding.

Disadvantages of Reusing Recycled PET Material

Despite its benefits, recycled PET may experience thermal and molecular degradation during repeated processing, leading to reduced strength and thermal stability. PET is also highly sensitive to moisture, and poor drying can cause hydrolytic degradation and processing defects. Higher recycled PET content may result in discoloration, reduced transparency, brittleness, and inconsistent product quality. Contamination from foreign materials can further affect processing performance, requiring additional cleaning and processing equipment, which increases operational costs and process complexity.

2.6.2 Textile and Fibre Applications

Characteristics of PET Used for Fibre Production

One of the largest and most common applications of recycled PET is polyester fibre production. Recycled PET fibres are extensively used in the textile industry for manufacturing clothing materials, carpets, upholstery fabrics, blankets, sportswear, industrial textiles, and pillow filling materials. Waste PET bottles and rejected PET products are collected, cleaned, shredded, and processed into flakes or pellets before being converted into polyester fibres through melting and extrusion processes.

PET used for fibre production is valued for its lightweight nature, durability, flexibility, and resistance to wear and tear. Polyester fibres produced from recycled PET also exhibit good tensile strength, abrasion resistance, wrinkle resistance, and moisture resistance, making them suitable for both domestic and industrial textile applications. In addition, PET fibres possess good thermal insulation properties and can maintain dimensional stability during use. These characteristics make recycled PET fibres suitable for products that require strength, comfort, and long service life.

Advantages of Using Recycled PET for Fibre Production

The use of recycled PET in fibre production offers significant environmental and economic benefits. It helps reduce plastic waste accumulation and environmental pollution by converting discarded PET materials into valuable textile products. This approach also reduces dependence on virgin polyester production, which lowers energy consumption and decreases the use of petroleum-based raw materials.

Another advantage is that fibre production can tolerate slightly lower material quality compared to food-grade PET applications, making it an effective recycling option for lower-grade PET waste. Recycled PET fibres are also relatively cost-effective and support sustainable manufacturing practices within the textile industry. Additionally, the recycling process promotes resource conservation and contributes to the development of circular economy systems.

Disadvantages of Using Recycled PET for Fibre Production

Despite its advantages, the use of recycled PET for fibre production also presents several challenges. Repeated thermal processing during recycling may cause polymer degradation, resulting in reduced fibre strength, durability, and overall material performance. Contamination from foreign materials, dyes, labels, and other plastics can negatively affect fibre quality, color consistency, and processing efficiency.

Another disadvantage is the potential release of microplastics during washing and usage of polyester-based textile products, which contributes to environmental pollution in water systems. Furthermore, recycled PET fibres may not always achieve the same quality and performance as virgin polyester fibres, especially in high-performance or specialized textile applications. Additional cleaning, sorting, and processing stages may also increase operational complexity and production costs.



2.6.3 Construction Applications

Research on sustainable construction materials has shown that recycled PET can be incorporated into plastic bricks, composite panels, geotextiles, insulation materials, lightweight construction products, and reinforced polymer composites. Waste PET materials are processed into flakes, fibres, or pellets and then blended with other construction materials to improve performance and reduce the use of conventional raw materials.

Characteristics of PET Used in Construction Applications

PET used in construction applications is valued for its lightweight nature, durability, corrosion resistance, and low moisture absorption properties. Recycled PET also possesses good tensile strength, chemical resistance, and thermal insulation characteristics, making it suitable for reinforcing construction materials and producing lightweight building products. In fibre-reinforced composites, PET contributes to improved crack resistance, flexibility, and impact strength. Its resistance to biological degradation and harsh environmental conditions also increases the durability of construction products.

Advantages of Using Recycled PET in Construction

The use of recycled PET in construction materials offers several environmental and economic advantages. It helps reduce plastic waste accumulation and landfill disposal by converting waste PET into useful building materials. Recycled PET products are generally lightweight, which reduces transportation and handling costs during construction activities. The incorporation of PET can also improve thermal insulation, crack resistance, and durability of certain construction products.

Another advantage is the reduction in the consumption of traditional construction materials such as natural aggregates and virgin polymers, supporting sustainable construction practices. In some applications, recycled PET composites also exhibit improved resistance to moisture, chemicals, and corrosion compared to conventional materials. Additionally, the approach supports circular economy initiatives and promotes resource conservation.

Disadvantages of Using Recycled PET in Construction

Despite its benefits, the use of recycled PET in construction applications also has limitations. One major disadvantage is the lower mechanical strength of some PET-based construction materials compared to traditional concrete or metal-based materials, which may limit their use in heavy structural applications. Poor bonding between PET particles and other construction materials can also affect product strength and durability.

Thermal degradation and exposure to ultraviolet radiation over long periods may reduce the performance of PET-based construction products. Contamination and inconsistent quality of recycled PET waste can further affect material properties and manufacturing consistency. In addition, specialized processing equipment and material treatment methods may increase production costs and operational complexity.

Another challenge is the limited long-term performance data available for some recycled PET construction materials, which may reduce confidence in their large-scale adoption within the construction industry.



2.6.4 3D Printing and Additive Manufacturing

Recent research has explored the use of recycled PET in additive manufacturing and 3D printing applications. Studies showed that recycled PET can be converted into 3D printing filament for fused deposition modelling (FDM) systems used in engineering prototyping and lightweight component production. Waste PET materials are cleaned, shredded, melted, and extruded into filament form for use in 3D printers.

Characteristics of PET Used in 3D Printing and Additive Manufacturing

PET used in 3D printing applications is valued for its good strength-to-weight ratio, durability, flexibility, and dimensional stability. Recycled PET also exhibits good layer adhesion, chemical resistance, and moisture resistance compared to some conventional 3D printing materials. Its lightweight nature and relatively high thermal stability make it suitable for producing engineering prototypes, lightweight components, and customized products. PET filaments can also provide smooth surface finishes and good impact resistance in printed parts.

Advantages of Using Recycled PET in 3D Printing

The use of recycled PET in additive manufacturing offers several environmental and economic benefits. It helps reduce plastic waste by converting discarded PET materials into useful 3D printing filament. This approach also lowers material costs for filament production and reduces dependence on virgin polymer materials.

Another advantage is the ability to produce customized components with minimal material waste compared to traditional manufacturing methods. Recycled PET filaments can be used for rapid prototyping, educational projects, product development, and lightweight engineering applications. Additionally, additive manufacturing supports sustainable production by enabling efficient material utilization and localized manufacturing processes.

Disadvantages of Using Recycled PET in 3D Printing

Despite its advantages, recycled PET used in 3D printing also presents several challenges. Repeated thermal processing during recycling and filament extrusion may degrade polymer properties, resulting in reduced strength, brittleness, and inconsistent print quality. Moisture absorption by PET can also negatively affect filament performance and lead to printing defects such as bubbles and poor layer bonding if proper drying is not performed.

Contamination and inconsistent particle quality in recycled PET waste may affect filament diameter consistency and printer performance. In some cases, recycled PET filaments may experience warping, nozzle clogging, or reduced dimensional accuracy during printing. Additional processing equipment and quality control systems are also required to produce reliable recycled PET filament, increasing production complexity and operational costs.



2.6.5 Energy Recovery

Energy recovery involves the incineration of PET and other plastic waste to generate heat or electricity. In this process, waste plastics are burned under controlled conditions, and the heat produced is used to generate steam for electricity production or industrial heating. Although this method reduces landfill waste, it also contributes to greenhouse gas emissions and results in the loss of recyclable material resources.

Characteristics of PET Used for Energy Recovery

PET possesses a relatively high calorific value, meaning it can release a significant amount of heat during combustion. This makes PET suitable for waste-to-energy applications. PET is also lightweight, combustible, and capable of producing stable heat energy during incineration. Due to its petroleum-based composition, PET can generate energy levels comparable to some conventional fossil fuels when properly processed in energy recovery systems.

Advantages of Energy Recovery from PET Waste

One major advantage of energy recovery is the significant reduction in waste volume sent to landfills. Studies show that incineration can reduce waste volume by up to 90% and waste mass by about 80–85%. ([Springer](#))

Energy recovery also allows the generation of electricity and heat from waste materials that may not be suitable for mechanical recycling due to contamination or degradation. Modern waste-to-energy plants can recover approximately 20–27% of energy as electricity in conventional systems, while combined heat and power systems can achieve overall energy recovery efficiencies above 60%. ([parliament.scot](#))

Another advantage is the reduction of dependence on landfill disposal and partial replacement of fossil fuel energy sources. Energy recovery can therefore contribute to waste management and energy generation simultaneously.

Disadvantages of Energy Recovery from PET Waste

Despite its benefits, energy recovery has several environmental disadvantages. Incineration of PET releases carbon dioxide and other greenhouse gases into the atmosphere, contributing to global warming. During combustion, a significant portion of the energy contained in PET waste is lost as heat to the atmosphere and surrounding systems. In electricity-only waste incineration plants, approximately 73–80% of the total energy may be lost during conversion processes, while only about 20–27% is recovered as usable electricity. ([parliament.scot](https://www.parliament.scot))

Another major disadvantage is the permanent loss of recyclable PET material that could otherwise be reused in manufacturing applications such as fibre production or new PET products. Incineration may also produce harmful emissions, ash residues, and air pollutants that require advanced emission control systems.

In addition, waste-to-energy facilities involve high installation and operational costs due to the need for specialized combustion, filtration, and pollution control equipment. Concerns over air pollution and greenhouse gas emissions continue to limit the sustainability of this approach compared to recycling-based waste management methods.



2.7 Simulation-Based Manufacturing Analysis

Modern manufacturing research increasingly integrates laboratory experimentation with simulation-based analysis. Banks et al. (2010) explained that manufacturing simulation improves industrial decision-making by enabling evaluation of production systems without disrupting actual industrial operations. Simulation tools are commonly used to analyze:

- Production throughput
- Machine utilization
- Material flow
- Waste generation
- Operational costs
- Resource utilization
- Manufacturing bottlenecks

Research studies involving recycled polymers demonstrated that simulation software can predict how recycled materials affect manufacturing performance and product quality.

Studies using AnyLogic simulation software showed that simulation models effectively evaluate manufacturing system behavior under different operating conditions and improve understanding of process optimization before industrial implementation.

Simulation-based manufacturing analysis therefore provides an important engineering approach for evaluating recycled PET material performance and operational efficiency.

2.8 Material Simulation Software

Modern engineering and materials science research increasingly relies on simulation software to analyze material behavior, optimize manufacturing systems, and predict product performance under different operating conditions.

Simulation software reduces experimental costs, improves process understanding, and enables researchers to evaluate different manufacturing scenarios before industrial implementation.

2.8.1 Finite Element Analysis (FEA) Software

Finite Element Analysis (FEA) software is widely used to simulate material behavior under thermal, structural, and mechanical loading conditions.

ANSYS : is commonly used for structural analysis, thermal analysis, stress analysis, and material property evaluation in polymer engineering and manufacturing research. Researchers use ANSYS to simulate stress distribution, deformation, heat transfer, and failure mechanisms in thermoplastic materials including recycled PET blends.

ABAQUS : is widely used for nonlinear material analysis, fracture mechanics, visco elastic analysis, and thermal degradation studies in polymer science. Research studies showed that ABAQUS provides accurate prediction of stress-strain behavior and deformation characteristics in recycled thermoplastic materials.

2.8.2 Polymer Processing Simulation Software

Polymer processing simulation software focuses on mould filling, cooling behavior, shrinkage, and defect prediction during injection moulding operations.

Autodesk Moldflow : is widely used to simulate injection moulding processes and predict defects .

SolidWorks Plastics : is used for analyzing mould filling patterns, cooling performance, shrinkage behavior, and moulding defects during thermoplastic processing. Researchers use the software to evaluate how recycled PET materials affect melt flow behavior and product quality.

2.8.3 Material Property Simulation Software

COMSOL Multiphysics : is used for coupled thermal, structural, fluid flow, and chemical analysis in engineering and materials science research. The software enables researchers to study heat transfer, polymer degradation, moisture diffusion, and thermal conductivity in recycled PET materials.

MATLAB : is widely used for statistical analysis, numerical modelling, material characterization, optimization, and predictive analysis in manufacturing research. Researchers use MATLAB to process experimental data and evaluate stress-strain relationships and thermal behavior in recycled polymer systems.

2.8.4 Molecular and Chemical Simulation Software

Materials Studio : is used for molecular modelling and simulation of polymers and advanced engineering materials. Researchers use the software to study molecular interactions, crystallinity, degradation mechanisms, and thermal behavior in recycled polymers.

LAMMPS : is an open-source molecular dynamics simulation software used for atomistic modelling of polymers and thermoplastic materials.

2.9 Research Gap

Several studies have examined PET recycling technologies, recycled polymer properties, sustainable manufacturing systems, and manufacturing simulation tools. However, limited research has specifically focused on:

- Recycling and re-purposing of rejected PET preforms generated during injection moulding operations
- Integration of recycled PET materials directly into manufacturing systems
- Combined use of laboratory experimentation and simulation-based manufacturing analysis
- Comparative evaluation of recycled PET blend configurations under manufacturing conditions
- Analysis of waste reduction, throughput improvement, material utilization, and operational cost reduction using recycled PET blends

Most previous studies focused primarily on post-consumer plastic recycling rather than industrial PET preform rejects generated during manufacturing operations.

Furthermore, limited research has investigated pulverized rejected PET preforms blended with virgin PET material at different recycling ratios while combining laboratory experimentation and manufacturing simulation analysis.

This study therefore seeks to address these research gaps by developing and evaluating a recycling and re-purposing strategy for rejected PET preforms generated during injection moulding operations using both experimental and simulation-based engineering approaches.

2.10 Chapter Summary

This chapter reviewed literature related to plastic manufacturing, PET material properties, injection moulding defects, PET recycling technologies, circular economy principles, sustainable manufacturing, recycled PET applications, material simulation software, and simulation-based manufacturing analysis.

The review established that PET waste generated during injection moulding operations possesses significant recycling potential and can contribute to improved manufacturing sustainability, reduced waste generation, improved resource utilization, and reduced operational costs.

The literature further showed that simulation software plays an important role in evaluating material behavior, manufacturing performance, and process optimization in recycled polymer systems.

Previous research studies demonstrated that recycled PET materials can be successfully reused in packaging, textiles, construction materials, automotive components, and engineering applications when proper recycling and processing conditions are maintained.

However, limited research has focused specifically on recycling rejected PET preforms generated during injection moulding operations using combined laboratory experimentation and simulation-based manufacturing analysis. This identified research gap forms the basis for the present study.

The next chapter presents the methodology adopted for evaluating the recycling and re-purposing of rejected PET preforms generated during injection moulding operations.

CHAPTER 3: METHODOLOGY

3.0 Introduction

This chapter outlines the research methodology adopted to achieve the objectives of the study on the recycling and re-purposing of rejected Polyethylene Terephthalate (PET) preforms generated during injection moulding operations. The methodology provides a systematic approach for identifying the causes and characteristics of PET preform rejects, evaluating the properties of recycled PET blends, determining suitable recycling applications, and analyzing the operational and economic effects of recycling within manufacturing processes.

The study employs a combination of qualitative and quantitative research methods to ensure a comprehensive evaluation of recycled PET material performance and manufacturing efficiency. Qualitative methods are used to investigate the causes of defects and operational practices through interviews, observations, and company records, while quantitative methods are applied in laboratory testing, simulation, and cost-performance analysis.

The chapter begins by presenting the research design adopted for the study, followed by the target population and sampling techniques. Data collection methods aligned with each research objective are then discussed in detail. The methodology further explains the experimental procedures used to produce recycled PET blends containing different percentages of pulverized rejected preforms mixed with virgin PET material. Simulation and comparative analyses are conducted on blends containing 10%, 15%, and 20% recycled PET content to evaluate mechanical, thermal, operational, and economic performance.

Additionally, the chapter outlines the data analysis techniques used to interpret both qualitative and quantitative findings. Measures implemented to ensure validity and reliability are discussed together with ethical considerations followed during the study.

3.1 Research Design

The research design provides the overall framework for conducting the study and ensuring that the research objectives are systematically addressed. This study adopts a mixed-methods research design incorporating descriptive, experimental, and comparative approaches to evaluate the recycling and re-purposing of rejected PET preforms in injection moulding operations.

A descriptive research design is employed to identify the major causes and characteristics of PET preform rejects generated during production. This involves collecting operational records, defect reports, production data, and observations from injection moulding processes. The descriptive approach enables the study to categorize common defects such as short shots, flash formation, bubbles, contamination, deformation, discoloration, and dimensional inconsistencies.

An experimental research design is used to evaluate the mechanical and thermal properties of recycled PET blends produced from rejected PET preforms. Rejected PET preforms are pulverized into powder form and blended with virgin PET material at different recycling ratios. Three recycled blend configurations are investigated:

- 10% recycled PET powder + 90% virgin PET
- 15% recycled PET powder + 85% virgin PET
- 20% recycled PET powder + 80% virgin PET

These blends are experimentally analyzed and simulated to determine the effects of recycling on material properties and manufacturing performance.

A comparative research design is further applied to compare the properties and operational performance of virgin PET material against the recycled PET blends. Key performance indicators such as tensile strength, thermal stability, production throughput, waste reduction, material utilization, and operational costs are analyzed across the different recycling ratios.

In addition, a simulation-based approach is adopted to model the behavior and performance of the recycled PET blends under different operating conditions. Engineering simulation tools and material property analysis techniques are used to evaluate the suitability of the recycled PET material for various industrial applications.

By integrating descriptive, experimental, comparative, and simulation-based approaches, the study ensures a comprehensive assessment of PET recycling within injection moulding operations.

3.2 Population and Sampling

3.2.1 Target Population

The target population for this study consists of personnel and operational systems involved in PET injection moulding and plastic manufacturing processes. The population includes:

- Production managers
- Quality control personnel
- Machine operators
- Process engineers
- Maintenance personnel
- Recycling specialists

The study also considers manufacturing records and defect databases from PET injection moulding companies and laboratory testing data and operational production records related to rejected PET preforms and recycled PET material.

3.2.2 Sampling Technique

A purposive sampling technique is employed to select manufacturing companies and personnel directly involved in PET preform production and recycling operations. This method ensures that only participants with relevant knowledge and experience contribute to the study.

Additionally, stratified sampling is used to categorize defect types and production records according to operational conditions, machine settings, and rejection causes. This allows for balanced representation of various defect categories and manufacturing conditions.

3.2.3 Sample Size

The sample size is determined based on the availability of production records, defect samples, and laboratory testing requirements. The study includes:

- Production and quality control records collected from PET injection moulding operations
- Samples of rejected PET preforms collected for recycling and testing
- Three recycled PET blend configurations (10%, 15%, and 20%)
- Multiple laboratory tests conducted for each blend configuration to ensure consistency and reliability
- Interviews are also conducted with selected production personnel and engineers to provide qualitative insights into defect causes and recycling practices.

3.2.4 Justification of Sampling Approach

Purposive sampling ensures that the study focuses on industries and personnel with direct involvement in PET injection moulding and recycling operations. Stratified sampling improves representation of different defect categories and operational conditions. The selected sample size provides sufficient data for meaningful experimental analysis, simulation, and comparison of recycled PET blends.

3.3 Data Collection Methods

This section outlines the data collection methods used to address each research objective. Both primary and secondary data sources are utilized to ensure accurate and reliable findings.

Data Collection for Objective : To identify the major causes and characteristics of PET preform rejects generated during injection moulding operations .Data for this objective is collected through the following methods:

- Company Production Records
- Quality Control Reports
- Interviews and Observations
- Defect Sample Collection

Data Collection for Objective 2: To experimentally evaluate the mechanical and thermal properties of recycled PET blends produced from rejected PET preforms. Experimental data is collected using the following methods:

- Material Preparation : Rejected PET preforms are cleaned, shredded, and pulverized into powder form before blending with virgin PET material
- Blend Formulation : Three recycled PET blend configurations are prepared
- Laboratory Testing : Mechanical and thermal tests are conducted on the recycled PET blends
- Simulation Data :Material property data collected from experiments is used in simulation software to evaluate material behavior under different manufacturing conditions.

Data Collection for Objective 3: To determine suitable recycling and re-purposing applications for recycled PET material .The following methods are used to collect data for this objective:

- Literature Review
- Comparative Material Analysis
- Expert Consultations

Data Collection for Objective 4: To analyze the effects of recycling on production throughput, waste reduction, material utilization, and operational costs .Data for this objective is collected through:

- Production Performance Records
- Cost Analysis Data
- Simulation Outputs
- Comparative Analysis

Performance indicators such as waste reduction percentage, material savings, and operational cost reduction are compared across the different recycling ratios.

Justification of Data Collection Methods

The integration of experimental testing, company records, interviews, simulations, and comparative analysis provides a comprehensive understanding of recycled PET material performance and manufacturing efficiency. Combining qualitative and quantitative methods enhances the reliability and accuracy of the study findings.

3.4 Data Analysis Techniques

Analysis for Objective 1: Identifying Causes and Characteristics of PET Preform Rejects

- Descriptive Statistical Analysis : Production defect data is analyzed using statistical tools such as Excel, SPSS, or Python to determine the frequency and distribution of defect types.
- Cause-and-Effect Analysis : Fish bone diagrams and Pareto analysis are used to identify the major causes contributing to PET preform rejection.
- Thematic Analysis : Interview responses are analyzed to identify recurring operational challenges associated with injection moulding defects.

Analysis for Objective 2: Evaluating Mechanical and Thermal Properties of Recycled PET Blends

- Comparative Experimental Analysis : Mechanical and thermal test results for the 10%, 15%, and 20% recycled PET blends are compared against virgin PET material.
- Statistical Analysis : Mean values, standard deviations, and percentage variations are calculated to evaluate changes in material properties.
- Simulation Analysis : Simulation tools are used to analyze material performance under manufacturing conditions and predict the effects of recycled content on product quality.

Analysis for Objective 3: Determining Suitable Recycling and Re-purposing Applications

- Material Suitability Analysis ; The experimental results are compared with industrial material standards to determine suitable applications for recycled PET blends.
- Comparative Evaluation : Each blend configuration is evaluated based on strength, thermal stability, durability, and processing characteristics.

Analysis for Objective 4: Analyzing Effects on Production Throughput, Waste Reduction, Material Utilization, and Operational Costs

- Cost-Benefit Analysis
- Throughput and Waste Analysis
- Comparative Performance Analysis

Justification of Data Analysis Methods

The combination of statistical, experimental, simulation-based, and comparative analysis techniques enables the study to comprehensively evaluate recycled PET material performance and operational efficiency.

3.5 Validity and Reliability

3.5.1 Validity

Content Validity : Research instruments and testing procedures are developed based on literature reviews, industrial standards, and expert consultation.

Construct Validity : Variables such as tensile strength, thermal stability, throughput, and cost reduction are clearly defined and measured using standardized procedures.

Internal Validity : Triangulation of data sources including company records, laboratory tests, interviews, and simulations improves accuracy and reduces bias.

External Validity : The findings are designed to be applicable to PET injection moulding operations within Zimbabwe and similar manufacturing environments.

3.5.2 Reliability

Pilot Testing : Testing procedures are first conducted on small sample batches to ensure consistency and accuracy.

Repeatability of Tests : Mechanical and thermal tests are repeated for each blend configuration to verify consistency of results.

Standardized Procedures : Uniform laboratory procedures and equipment settings are used throughout the study.

Data Verification : Company records and experimental results are cross-checked to ensure reliability of findings.

By implementing these measures, the study ensures credible and reproducible research results.

3.6 Ethical Considerations

Informed Consent : Participants involved in interviews are informed about the purpose of the study and participate voluntarily.

Confidentiality and Anonymity : Company records and participant information are treated confidentially and used strictly for academic purposes.

Voluntary Participation : Participants are allowed to withdraw from the study at any stage without penalty.

Responsible Data Handling : Collected data is securely stored in password-protected systems and only accessible to authorized researchers.

Ethical Approval : Permission is obtained from relevant institutions and manufacturing companies before conducting data collection and laboratory testing.

By adhering to ethical standards, the study maintains integrity, transparency, and credibility.

3.7 Chapter Summary

This chapter presented the methodology adopted for investigating the recycling and re-purposing of rejected PET preforms generated during injection moulding operations. The study employed a mixed-methods approach incorporating descriptive, experimental, comparative, and simulation-based research designs.

The chapter discussed the target population, sampling techniques, and data collection methods used to gather operational records, defect samples, laboratory test results, and expert opinions. Experimental analysis was conducted on recycled PET blends containing 10%, 15%, and 20% pulverized rejected PET material mixed with virgin PET material.

Various data analysis techniques including statistical analysis, comparative evaluation, simulation analysis, and cost-benefit analysis were outlined. Measures implemented to ensure validity, reliability, and ethical compliance were also discussed.

CHAPTER 4: RESULTS

4.0 Introduction

This chapter presents the simulation results and material performance evaluation obtained from the thermo-mechanical analysis of virgin Polyethylene Terephthalate (PET) and recycled PET (rPET) blends during the stretch blow moulding process. The chapter focuses on the behavior of PET blends containing 10%, 15%, and 20% recycled PET material under industrial processing conditions using COMSOL Multiphysics .

The results presented in this chapter were generated using coupled heat transfer and solid mechanics simulations to evaluate thermal behavior, deformation characteristics, stress distribution, material stiffness, and structural performance during preform reheating and bottle formation. Material properties used in the simulation model were derived from the rule of mixtures and engineering property estimation techniques.

The chapter further presents comparative analysis of the recycled PET blends in terms of mechanical performance, thermal response, deformation stability, and manufacturability. Simulation outputs obtained from COMSOL Multiphysics were used to determine the most suitable recycled PET blend ratio capable of maintaining acceptable bottle quality while improving sustainability and reducing raw material consumption. The results obtained from the simulations provide an engineering basis for evaluating the feasibility of integrating recycled PET material into industrial PET bottle manufacturing systems.

4.1 Thermo-Mechanical Material Properties

The first stage of the analysis involved determining the thermo-mechanical properties of the virgin PET material and recycled PET blends. The incorporation of recycled PET material resulted in gradual changes in the material properties due to thermal degradation, molecular chain scission, and reduced molecular weight caused by repeated processing cycles.

This research utilizes the Linear Rule of Mixtures (ROM) formula as a deterministic model to interpolate material property fields across the design configurations. The ROM formulation represents property behavior as a function of component volume or mass fractions:

$$P_{blend} = V_v \times P_v + V_r \times P_r$$

Where P_{blend} represents the calculated blend property field, P_v and P_r denote the specific baseline property values of the Pure Virgin and Pure Recycled components, and V_v and V_r denote their respective volume fractions inside the structural matrix (where $V_v + V_r = 1.0$). The resulting calculated cards establish the direct physical inputs for structural finite element analysis validation and multi-physics coupling.

The calculated material properties showed that increasing recycled PET content reduced stiffness, tensile strength, and thermal stability while slightly increasing thermal expansion behavior. These changes directly influenced the deformation and structural response of the preforms during reheating and blow moulding

4.1.1 Material Property Results

The thermo-mechanical material properties generated for the different PET blend configurations are summarized below.

Property	0%rPET	10% rPET	15% rPET	20% rPET	100% rPET
Elastic Modulus (N/m ²)	3,000,000,000	2,945,000,000	2,917,500,000	2,890,000,000	2,450,000,000
Poisson's Ratio	0.394	0.396	0.396	0.397	0.410
Shear Modulus (N/m ²)	1,076,000,000	1,055,280,000	1,044,920,000	1,034,560,000	868,800,000
Mass Density (kg/m ³)	1,380.0	1,376.0	1,374.0	1,372.0	1,340.0
Tensile Strength (N/m ²)	60,000,000	58,200,000	57,300,000	56,400,000	42,000,000
Yield Strength (N/m ²)	55,000,000	53,300,000	52,450,000	51,600,000	38,000,000
Thermal Expansion Coeff. (/K)	0.0000700	0.0000715	0.0000723	0.0000730	0.0000850
Thermal Conductivity [W/(m·K)]	0.2400	0.2375	0.2363	0.2350	0.2150
Specific Heat [J/(kg·K)]	1,200.0	1,211.0	1,216.5	1,222.0	1,310.0

The results showed that Young's modulus gradually decreased as recycled PET content increased. Virgin PET exhibited the highest stiffness while the 20% recycled PET blend showed the lowest stiffness. This reduction occurred because recycled PET experiences polymer chain degradation during previous thermal processing cycles.

The tensile strength and yield strength also decreased with increasing recycled PET content. However, the reduction observed in the 10% and 15% recycled PET blends remained relatively small, indicating that moderate recycled PET incorporation can still maintain acceptable mechanical performance for industrial bottle production applications. Thermal conductivity slightly decreased with increasing recycled PET content, indicating a minor reduction in heat transfer efficiency during reheating. The density of the material also showed a slight decrease as recycled PET content increased.

4.2 Simulation Procedure

The simulation procedure was divided into two sequential stages: preform reheating simulation followed by blow moulding simulation. This approach was selected to closely replicate the actual industrial stretch blow moulding process used in PET bottle manufacturing.

The reheating simulation was first conducted to evaluate thermal distribution and deformation behavior of the PET preforms during infrared heating. The preforms were heated within the PET rubbery processing region at an average temperature of 110°C (383.15 K). This stage was important because temperature distribution before inflation directly affects wall thickness uniformity, stretching behavior, and final bottle quality. The reheating analysis further enabled evaluation of how recycled PET content influences thermal conductivity, thermal expansion, and structural deformation during heating.

After reheating, the blow moulding simulation was performed using the thermal conditions generated from the first stage. During this stage, the preforms were subjected to industrial blow moulding conditions including internal inflation pressure, axial stretching forces, and cooling effects. An internal pressure of 3.5 MPa (3.5×10^6 Pa) was applied to simulate bottle inflation, while mould wall temperatures were maintained between 15°C and 20°C (288.15 K – 293.15 K) to represent industrial cooling conditions. Gravity loading of 9.81 m/s² was also included in the model.

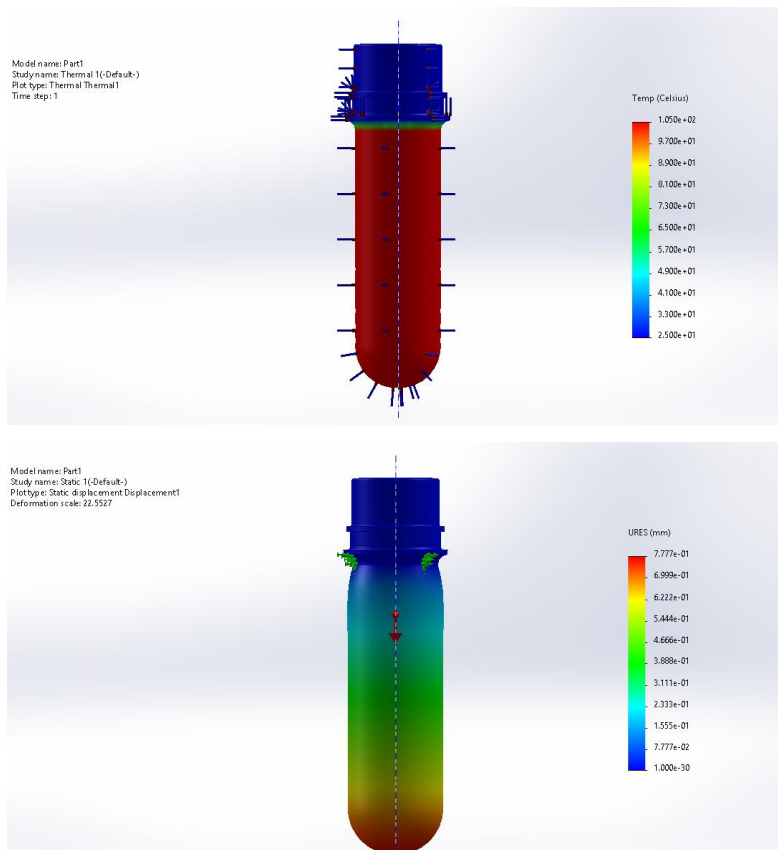
The coupled thermo-mechanical simulation procedure allowed realistic analysis of temperature distribution, deformation behavior, stress development, wall thickness variation, and structural performance of virgin PET and recycled PET blends under actual manufacturing conditions.

4.3 Preform Reheating Simulation Results

The reheating simulations were conducted to evaluate thermal distribution and deformation behavior of the PET preforms during the infrared heating stage of stretch blow moulding. The simulations were performed at industrial reheating temperatures within the rubbery processing region of PET. The results obtained from COMSOL Multiphysics showed that the thermal response of the material was significantly influenced by recycled PET content.

4.3.1 10% Recycled PET Blend

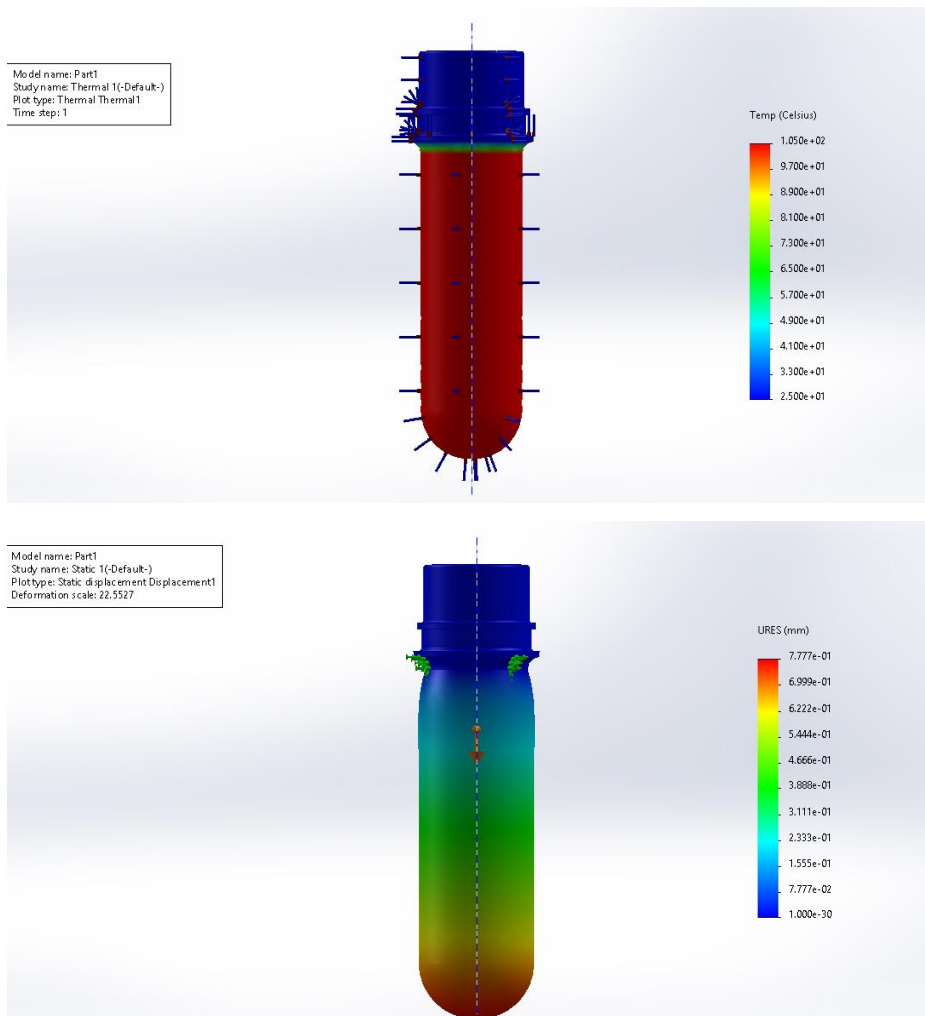
The 10% recycled PET blend demonstrated stable thermal behavior and uniform temperature distribution during reheating. The preform maintained good structural stability with minimal excessive deformation under thermal loading conditions. The simulation results indicated that the 10% blend retained high stiffness and sufficient resistance to deformation during heating. Heat distribution across the preform wall was relatively uniform, resulting in stable reheating .



4.3.2 15% Recycled PET Blend

The 15% recycled PET blend also demonstrated stable reheating behavior with acceptable thermal uniformity and controlled deformation characteristics. Compared to the 10% blend, a slight increase in thermal deformation was observed due to reduced material stiffness.

However, the deformation remained within acceptable industrial processing limits. The material maintained sufficient structural integrity during reheating and demonstrated good process stability during thermal loading. The simulation further showed that the 15% blend achieved a balance between sustainability improvement and acceptable mechanical performance.

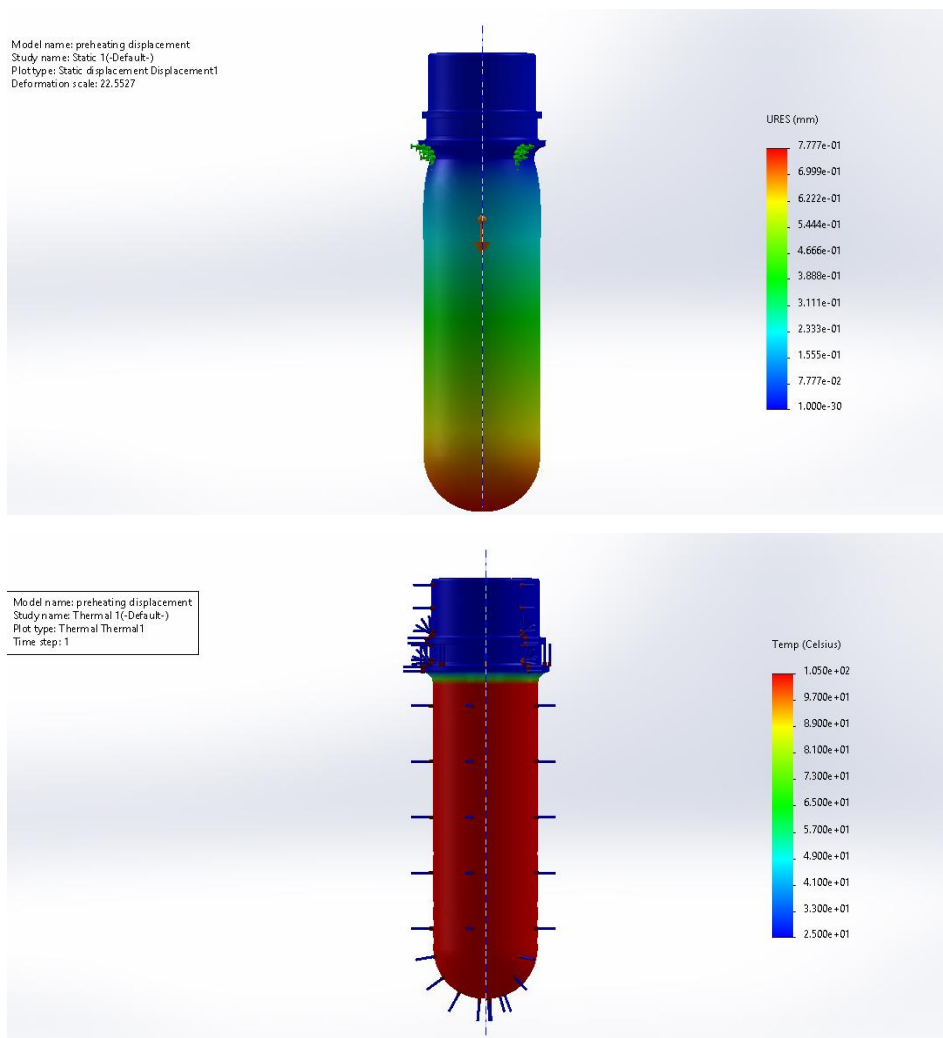


4.3.3 20% Recycled PET Blend

The 20% recycled PET blend exhibited the highest thermal deformation during reheating. The reduction in stiffness caused the material to experience increased deformation and localized instability under heating conditions.

The simulation showed increased thermal sensitivity and reduced structural resistance compared to the 10% and 15% blends. Non-uniform deformation patterns were also observed, indicating a greater risk of uneven wall thickness during bottle formation.

Although the material remained processable, the results suggested that higher recycled PET incorporation may negatively affect process consistency and final bottle quality.



4.4 Blow Moulding Simulation Results

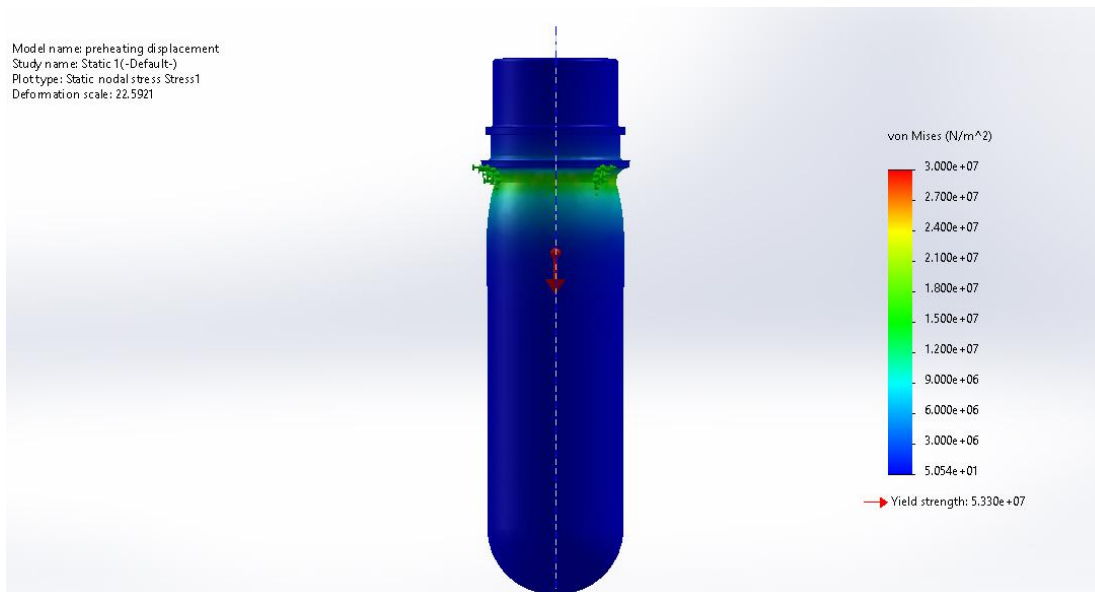
The blow moulding simulations were performed to analyze bottle inflation behavior, stress development, wall thickness distribution, and structural stability during expansion of the PET preforms. The results showed that all PET blend configurations were capable of undergoing inflation and bottle formation. However, deformation quality and structural stability varied depending on recycled PET content.

4.4.1 Stress Distribution Results

The Von Mises stress distribution simulations were conducted to evaluate the structural response of the PET preforms and bottles during biaxial stretching and inflation under industrial blow moulding conditions. The stress distribution patterns obtained from COMSOL Multiphysics provided insight into the effect of recycled PET content on deformation resistance, stress concentration, and structural stability during bottle formation.

The results showed that stress concentration increased progressively with increasing recycled PET content due to reductions in Young's modulus, shear modulus, and ultimate strain caused by polymer degradation and molecular chain scission in the recycled material. As the recycled PET ratio increased, the material exhibited lower resistance to deformation, resulting in higher localized stresses during inflation and stretching.

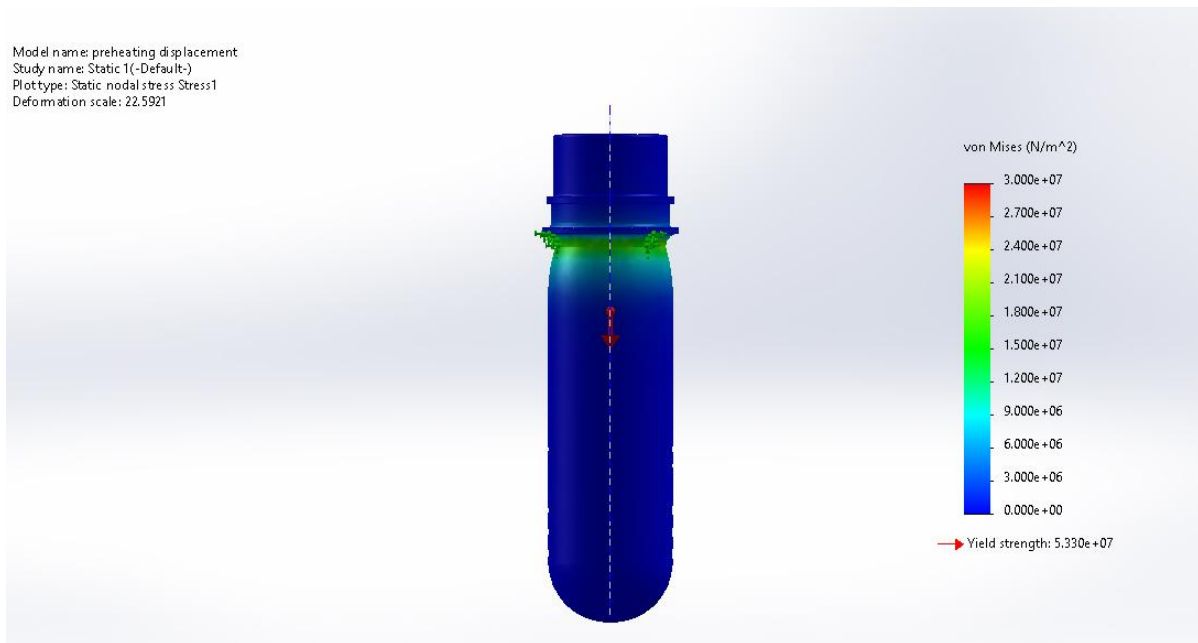
10_%rPET_(Von_Mises)



For the 10% recycled PET blend, the Von Mises stress distribution remained relatively uniform throughout the bottle structure. The stress contours indicated smooth stress transfer across the bottle wall with minimal localized stress concentration regions. The highest stress regions were mainly concentrated around the bottle shoulder and base transition zones, which are typical high-deformation areas during stretch blow moulding. However, the stress magnitudes remained within acceptable structural limits, indicating good mechanical stability and sufficient resistance to biaxial stretching. The relatively uniform stress distribution suggested stable material orientation and controlled deformation during bottle expansion.

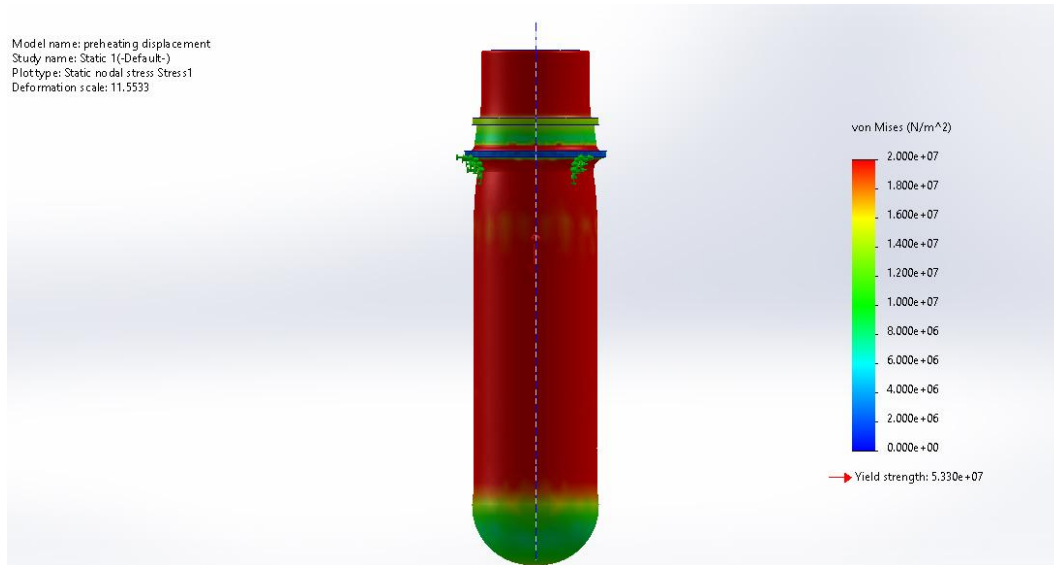
15_%rPET_(Von_Mises)

Model name: preheating displacement
Study name: Static 1(-Default-)
Plottype: Static nodal stress Stress1
Deformation scale: 22.5921



The 15% recycled PET blend also demonstrated stable structural behavior, although slightly higher stress concentrations were observed compared to the 10% blend. Increased stress localization was visible around the bottle shoulder region and near the base corners due to the reduction in material stiffness and deformation resistance. Despite this increase, the stress distribution remained relatively balanced throughout the bottle structure, and no excessive localized overstressing was observed. The simulation results indicated that the 15% blend maintained acceptable structural integrity and deformation control during inflation. The moderate increase in stress concentration suggested that the material still possessed sufficient molecular orientation and load-carrying capability for industrial bottle manufacturing applications.

20_%rPET_(Von_Mises)



The 20% recycled PET blend exhibited the highest Von Mises stress levels among all blend configurations. Significant stress concentration regions developed around the bottle shoulder, neck transition, and lower base regions during inflation. The stress contours showed more pronounced localized stress accumulation caused by reduced stiffness, lower ultimate strain capacity, and increased thermal deformation behavior of the material. The higher stress magnitudes indicated reduced resistance to biaxial stretching and increased susceptibility to localized thinning and structural instability. In some regions, the stress patterns suggested a higher probability of material weakening and non-uniform wall thickness formation during industrial processing conditions.

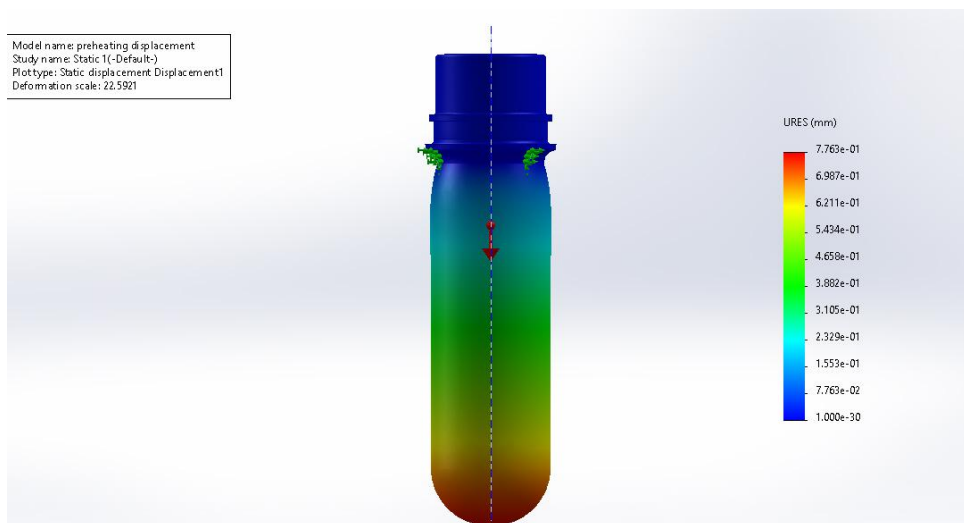
4.4.2 Wall Thickness Distribution

Wall Thickness Distribution and Displacement Behavior

Wall thickness distribution plays a major role in determining bottle quality, mechanical strength, pressure resistance, and structural durability in PET bottle manufacturing. Although the COMSOL Multiphysics analysis was primarily conducted using displacement and deformation simulations, the displacement behavior of the preform during inflation directly relates to the final wall thickness distribution of the bottle.

During stretch blow moulding, regions experiencing higher displacement and excessive stretching generally become thinner because the material is stretched over a larger surface area. Conversely, regions with controlled and uniform displacement tend to maintain more uniform wall thickness distribution. The displacement simulation therefore provided an indirect but reliable indication of potential wall thickness variation throughout the bottle structure.

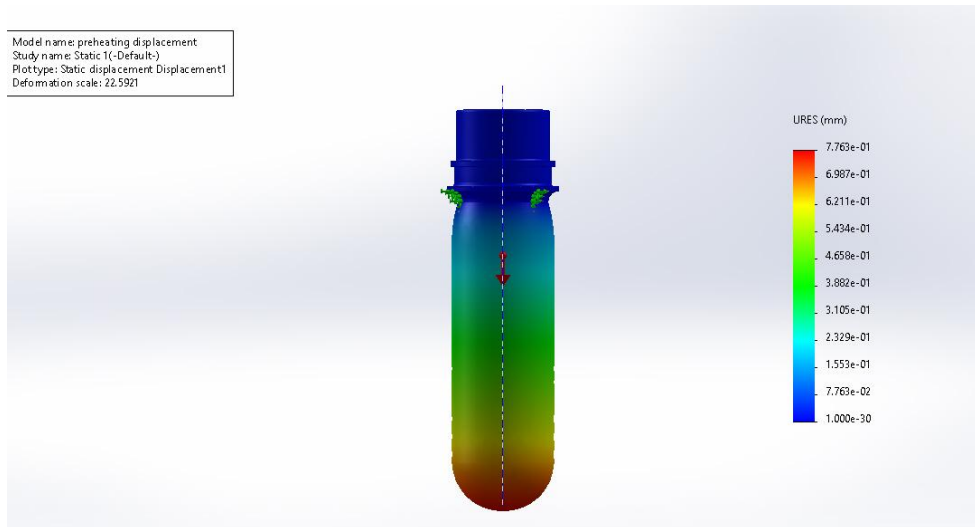
10_%rPET_(Displacement)



The simulation results showed that the 10% recycled PET blend exhibited relatively uniform displacement patterns during inflation. The material expanded consistently across the bottle wall with controlled deformation behavior and limited localized overstretching. This indicated that the

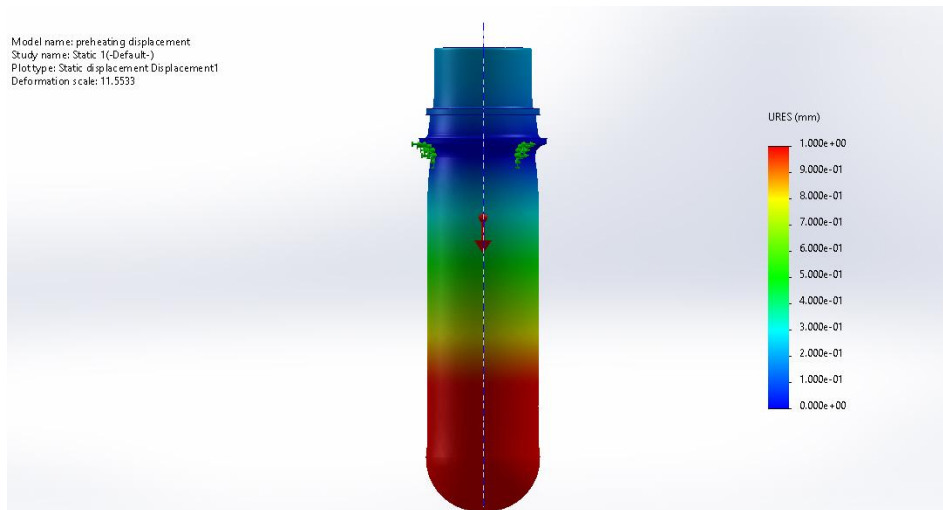
final bottle would likely achieve relatively uniform wall thickness distribution with minimal thin sections. The stable displacement behavior further suggested good molecular orientation and balanced material flow during biaxial stretching

15_%rPET_(Displacement)



The 15% recycled PET blend also demonstrated acceptable displacement distribution during inflation. Slightly higher displacement values were observed compared to the 10% blend due to the moderate reduction in stiffness and deformation resistance. However, the deformation remained relatively balanced throughout the bottle body, indicating that wall thickness variation would remain within acceptable industrial limits. The simulation suggested that the 15% blend could still maintain adequate bottle strength and structural stability while incorporating a larger proportion of recycled material.

20_%rPET_(Displacement)



The 20% recycled PET blend exhibited the highest displacement magnitudes during bottle inflation. Significant localized deformation occurred particularly around the bottle shoulder, sidewall transition regions, and lower base areas. These regions experienced excessive stretching due to reduced stiffness and lower resistance to deformation. The increased displacement indicated a greater probability of localized wall thinning during bottle formation.

Excessive displacement and non-uniform deformation patterns suggested that some bottle regions would become thinner than others, creating potential weak points within the structure. Such thinning may reduce pressure resistance, structural rigidity, and durability of the final bottle during handling and service conditions. The results therefore indicated that higher recycled PET incorporation increases the risk of uneven wall thickness distribution and reduced bottle quality.

4.5 Chapter Conclusion

This chapter presented the thermo-mechanical simulation results and material performance evaluation of virgin PET and recycled PET blends during the stretch blow moulding process using COMSOL Multiphysics. The analysis focused on evaluating the influence of recycled PET incorporation on thermal behavior, deformation characteristics, stress distribution, displacement behavior, and structural stability under industrial processing conditions.

The simulation results showed that increasing recycled PET content gradually reduced material stiffness, thermal stability, and resistance to deformation due to polymer degradation and molecular chain scission associated with repeated thermal processing. The reheating simulations demonstrated that recycled PET content significantly influenced temperature distribution and thermal deformation behavior during infrared heating of the preforms.

The blow moulding simulations further showed that increasing recycled PET content increased displacement and Von Mises stress concentration during bottle inflation. The 10% recycled PET blend demonstrated the most stable deformation behavior, lowest stress concentration, and best potential wall thickness uniformity. The 20% recycled PET blend exhibited excessive displacement, increased localized stress concentration, and a higher likelihood of wall thinning and structural instability during bottle formation.

The 15% recycled PET blend demonstrated balanced thermo-mechanical performance by maintaining acceptable stress distribution, controlled displacement behavior, and relatively uniform wall thickness characteristics while still incorporating a significant proportion of recycled material. The blend therefore provided a suitable compromise between structural performance, process stability, sustainability improvement, and recycled material utilization.

Overall, the simulation results confirmed that recycled PET material generated from rejected PET preforms can successfully be reintegrated into PET bottle manufacturing systems when controlled blend ratios are used. The study further demonstrated the effectiveness of COMSOL Multiphysics as a tool for evaluating the thermo-mechanical behavior of recycled polymer blends under realistic industrial stretch blow moulding conditions.

CHAPTER 5: ANALYSIS

5.0 Introduction

This chapter presents the detailed analysis and interpretation of the thermo-mechanical simulation results obtained from the evaluation of virgin Polyethylene Terephthalate (PET) and recycled PET (rPET) blends during the stretch blow moulding process. The chapter focuses on assessing the influence of recycled PET incorporation on material behavior, structural performance, thermal response, deformation characteristics, manufacturing feasibility, and economic performance under industrial processing conditions.

The analysis presented in this chapter is based on the coupled heat transfer and solid mechanics simulations conducted using COMSOL Multiphysics. The simulation results generated during the reheating and blow moulding stages were analyzed to determine how different recycled PET blend ratios affect temperature distribution, stress development, displacement behavior, wall thickness uniformity, and overall bottle quality during production. Comparative analysis was conducted for the 10%, 15%, and 20% recycled PET blends in order to identify the most suitable blend configuration for sustainable PET bottle manufacturing applications.

The chapter further evaluates the relationship between recycled PET content and thermo-mechanical material properties such as Young's modulus, thermal conductivity, tensile strength, thermal deformation behavior, and structural stability. The influence of polymer degradation and molecular chain scission on material performance is also discussed in relation to the observed simulation behavior. Particular attention is given to the effect of recycled PET content on stress concentration, localized deformation, and potential wall thinning during bottle inflation.

Based on the combined thermo-mechanical, structural, thermal, and economic analyses, the chapter identifies the optimum recycled PET blend ratio capable of achieving acceptable bottle quality while improving sustainability and reducing raw material costs. The chapter concludes by presenting the overall conclusions of the study together with recommendations for industrial implementation and future research on recycled PET applications in stretch blow moulding operations.

5.1 Material Performance Analysis

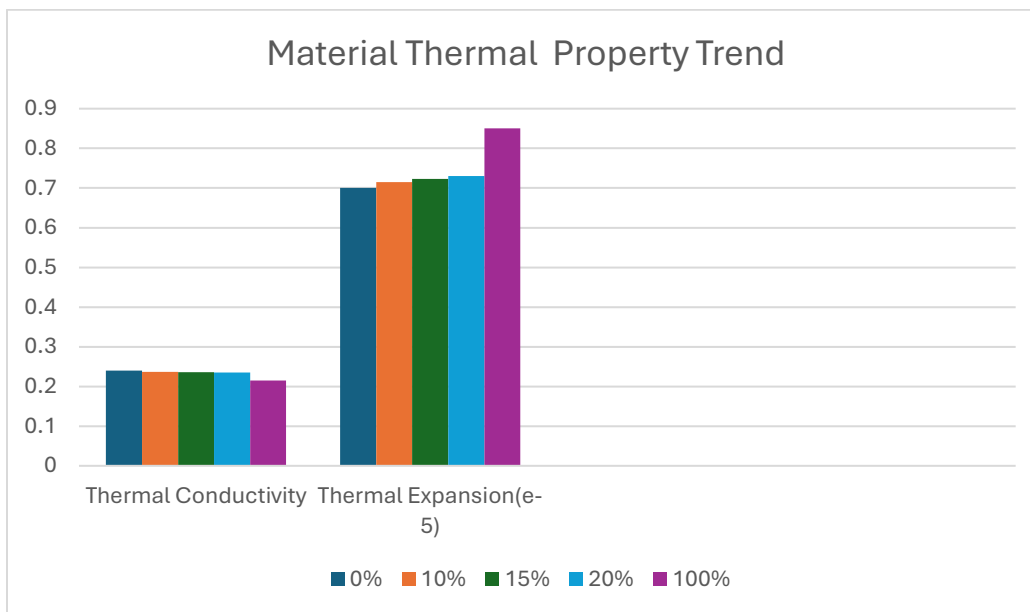
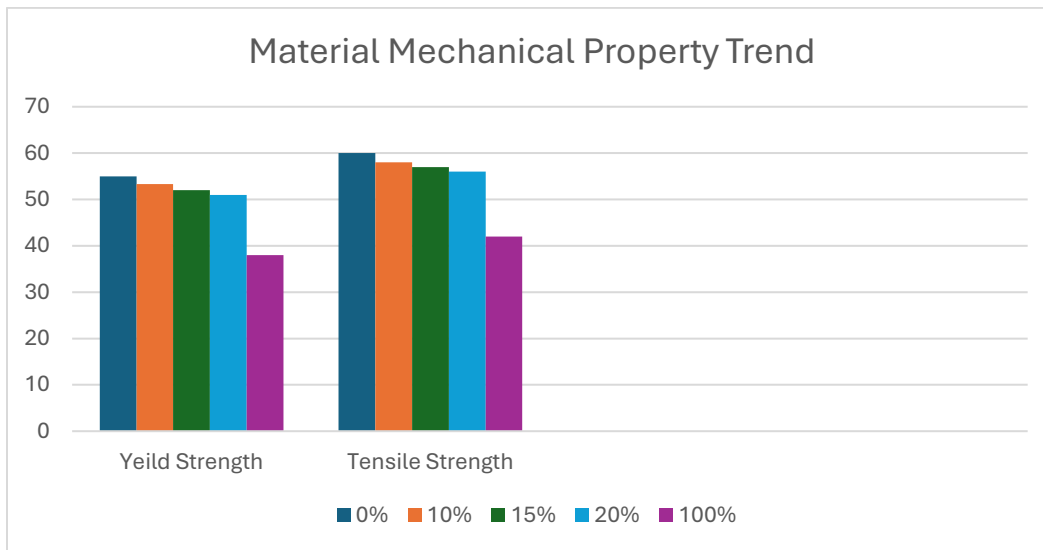
The mechanical and thermal property analysis was conducted to evaluate the suitability of recycled PET blends for industrial stretch blow moulding applications. The results showed that increasing recycled PET content gradually reduced the structural stiffness and thermal stability of the material due to polymer degradation and molecular chain scission caused by repeated thermal processing cycles.

One of the most significant mechanical properties analyzed was the elastic modulus, which represents the stiffness and resistance of the material to deformation during bottle inflation. Virgin PET exhibited the highest elastic modulus of $3.00 \times 10^9 \text{ N/m}^2$, while the 10%, 15%, and 20% recycled PET blends showed values of $2.945 \times 10^9 \text{ N/m}^2$, $2.9175 \times 10^9 \text{ N/m}^2$, and $2.89 \times 10^9 \text{ N/m}^2$ respectively. The gradual reduction in elastic modulus indicated that increasing recycled PET content reduced the resistance of the material to stretching and deformation during blow moulding. This behavior corresponded with the increased displacement and wall thinning observed in the simulation results for higher recycled PET blends.

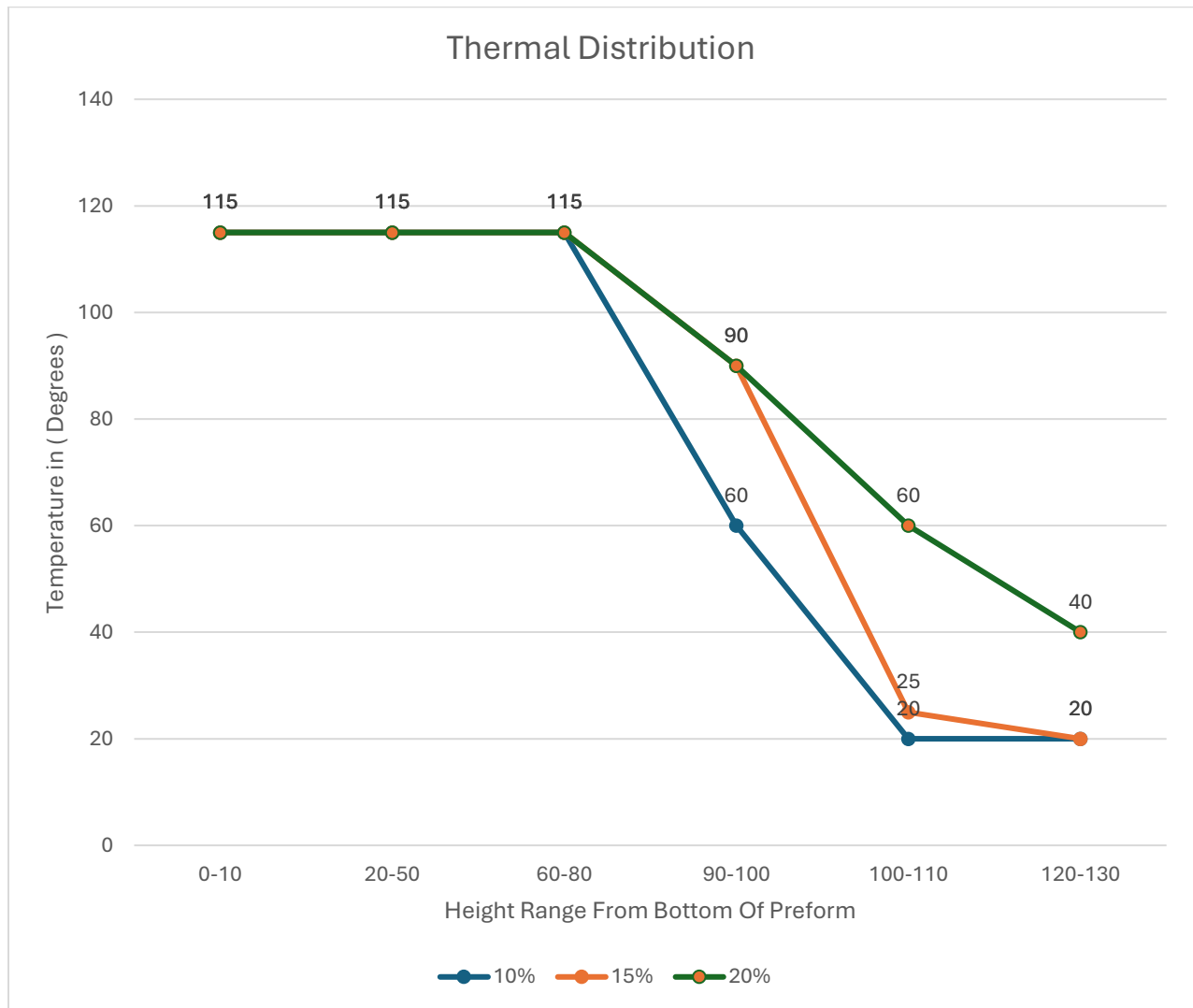
Tensile strength and yield strength also decreased progressively with increasing recycled PET content. Virgin PET exhibited a tensile strength of $6.0 \times 10^7 \text{ N/m}^2$ and yield strength of $5.5 \times 10^7 \text{ N/m}^2$. The 15% recycled PET blend maintained acceptable values of $5.73 \times 10^7 \text{ N/m}^2$ and $5.245 \times 10^7 \text{ N/m}^2$ respectively, indicating that the material still possessed sufficient load-carrying capability and structural stability for bottle formation. However, the 20% recycled PET blend showed greater reductions in strength, increasing the possibility of excessive deformation and localized structural weakening during inflation.

The thermal properties also showed gradual variation with increasing recycled PET incorporation. Thermal conductivity decreased from $0.2400 \text{ W/(m}\cdot\text{K)}$ for virgin PET to $0.2363 \text{ W/(m}\cdot\text{K)}$ and $0.2350 \text{ W/(m}\cdot\text{K)}$ for the 15% and 20% recycled PET blends respectively. Reduced thermal conductivity indicated lower heat transfer efficiency during infrared reheating, which may contribute to non-uniform temperature distribution during processing.

The thermal expansion coefficient increased from $7.0 \times 10^{-5} /K$ for virgin PET to $7.23 \times 10^{-5} /K$ and $7.30 \times 10^{-5} /K$ for the 15% and 20% blends respectively. Increased thermal expansion behavior contributed to the higher displacement and deformation observed during the blow moulding simulations. Similarly, the specific heat capacity increased gradually with increasing recycled PET content, indicating that higher recycled PET blends required slightly more thermal energy during reheating.



5.2 Thermal Analysis



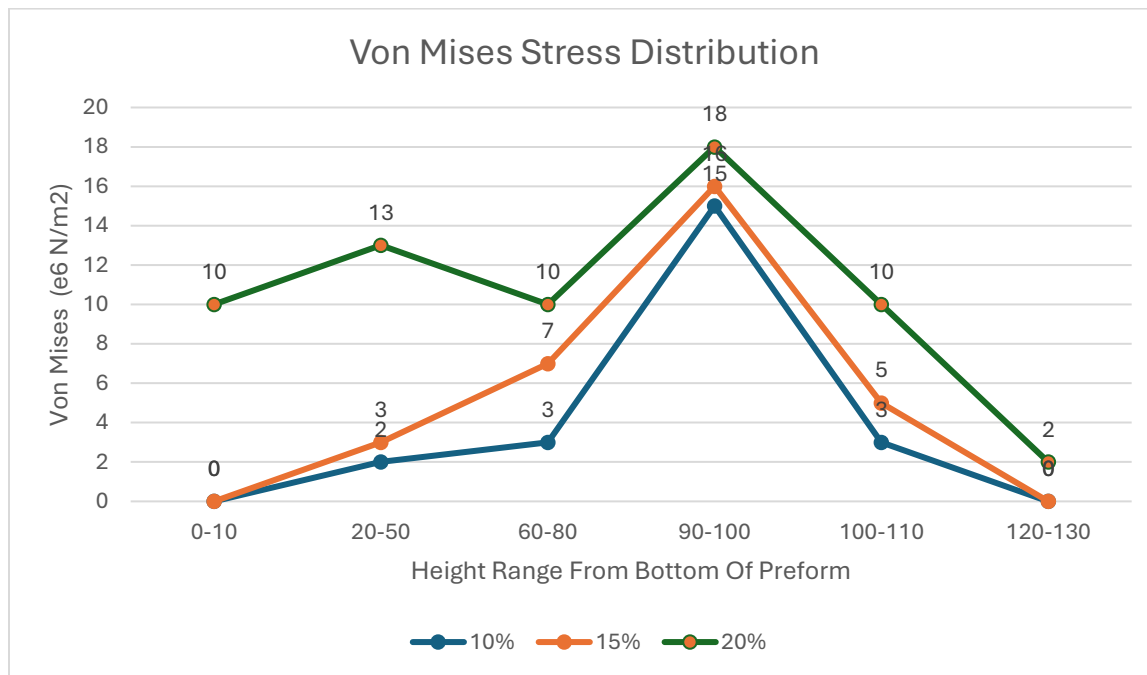
The thermal processing architecture demonstrates a tightly controlled, uniform heating profile along the main structural body of the preform, spanning from 0 millimeters up to 80 millimeters. Within this continuous lower zone, all three material compositions exhibit an identical equilibrium processing temperature maintained precisely at 115 degrees Celsius. This steady thermal state ensures that the polymer matrix remains sufficiently compliant above its glass transition temperature to facilitate clean material redistribution during stretch blow molding.

However, severe thermal divergence takes place immediately upon entering the upper transition and neck areas, spanning from 90 millimeters to 130 millimeters. In this region, localized cooling mechanisms produce distinct thermodynamic paths based on rPET percentage. The baseline 10% blend undergoes rapid and aggressive cooling, falling sharply to 60 degrees Celsius in the 90-to-100-millimeter range, and flattening out to a cold 20 degrees Celsius at the terminal neck boundary.

The mid-tier 15% blend showcases a slightly delayed cooling response, recording a moderate 90 degrees Celsius at the transition zone before settling to 25 degrees and eventually matching the 20 degrees Celsius baseline at the absolute tip. In stark contrast, the high-concentration 20% rPET blend retains an extreme thermal load throughout the upper column. It maintains a high temperature of 90 degrees Celsius at the transition point, resists intermediate cooling by hanging at 60 degrees Celsius at 100 to 110 millimeters, and terminates at an elevated 40 degrees Celsius in the 120-to-130-millimeter neck zone, indicating localized heat retention and poor thermal dissipation

5.3 Stress and Structural Analysis

The structural analysis results indicated that increasing recycled PET content increased localized stress development during bottle inflation. Excessive stress concentration increases the risk of material thinning and structural failure. The 15% recycled PET blend maintained acceptable stress distribution while still achieving improved sustainability performance compared to the 10% blend.



The resulting Von Mises stress distribution directly mirrors the underlying thermal patterns, revealing a compounding structural penalty as the percentage of recycled polymer escalates. Throughout the primary preform body between 0 and 80 millimeters, the internal residual stresses for both the 10% and 15% configurations remain safely suppressed, fluctuating within a tight, acceptable range of 0 to 7 megapascals. Conversely, the 20% rPET matrix acts as a major mechanical outlier, exhibiting significantly elevated internal stresses ranging between 10 and 13 megapascals in the exact same zone, despite experiencing identical external thermal parameters. This indicates that higher recycled fractions significantly alter melt rheology, introducing

structural micro-impurities, un-melted crystalline nodes, or altered molecular weight distributions that inherently resist uniform fluid flow and lock in high base-level stresses.

The definitive stress characteristic across all three material blends is an abrupt, massive stress spike localized precisely within the 90-to-100-millimeter spatial window. Here, the internal stresses climb violently to reach global peaks of 15 megapascals for the 10% blend, 15 megapascals for the 15% blend, and a highly severe 18 megapascals for the 20% blend. This localized stress concentration is caused by severe spatial thermal shock. Because the preform transitions sharply from a hot, compliant 115 degrees Celsius zone down to an actively cooled zone over a narrow physical distance, severe differential volumetric contraction occurs between adjacent polymer layers. As the outer skin layer contracts rapidly against a warmer, slower-moving core, extreme shear stresses are locked into the solidifying material matrix at this exact coordinate.

5.4 Industrial Risk Mitigation and Process Control Interventions

Correlating these empirical findings against established industrial production standards reveals a critical manufacturing risk regarding neck deformation. Manufacturing specifications dictate that the preform neck area must maintain a strict maximum temperature threshold of 20 degrees Celsius during handling to guarantee absolute dimensional stability and avoid thread warpage. While the 10% and 15% rPET blends successfully achieve this target profile, the 20% blend completely violates the constraint by retaining a hot 40 degrees Celsius temperature at the 120-to-130-millimeter level, while simultaneously carrying a lingering 2 megapascal residual structural stress.

Operating a 20% recycled blend under standard production parameters will inevitably result in severe mechanical failures, characterized by sagging necks, ovality defects in the container finish, and thread distortion during the high-pressure blow-molding sequence. To mitigate this threat and allow for successful high-recycled-content manufacturing, the industrial process must be modified. Engineering interventions should include deploying localized, high-velocity chilled air or targeted cooling rings directly focused onto the upper preform segments between 90 and 130 millimeters. Artificially forcing the thermal profile of the 20% blend down to meet the mandatory 20 degrees Celsius constraint will safely minimize localized stress concentrations, eliminate dimensional deformation, and ensure product quality without sacrificing the sustainability benefits of high-recycled-content materials.

5.5 Cost Analysis

The incorporation of recycled PET material reduces dependence on virgin PET resin and lowers raw material costs during production. Since rejected PET preforms are reprocessed and reused within the manufacturing system, waste disposal costs are also reduced.

The 10% recycled PET blend achieved moderate material savings but provided lower sustainability benefits. The 20% blend achieved the highest reduction in virgin material consumption but may increase operational risks due to reduced process stability and increased quality control requirements.

The 15% recycled PET blend demonstrated the most economically balanced configuration because it achieved significant virgin material reduction while maintaining acceptable process stability and bottle quality.

5.6 Selection of the Optimum Blend Ratio

Based on the thermo-mechanical simulation results, material performance analysis, thermal behavior evaluation, and cost analysis, the 15% recycled PET blend was selected as the optimum blend ratio for industrial PET bottle manufacturing applications.

The 15% blend was selected because it demonstrated:

- Acceptable mechanical strength
- Stable thermal behavior
- Controlled deformation during reheating
- Uniform wall thickness distribution
- Acceptable stress distribution
- Improved sustainability performance
- Significant virgin material reduction
- Reduced manufacturing waste
- Better economic balance between performance and recycling

Although the 10% blend exhibited slightly better mechanical performance, the sustainability improvement achieved was lower. The 20% blend achieved higher recycled material utilization but demonstrated increased thermal instability and structural weakening.

The 15% blend therefore provided the best compromise between engineering performance, sustainability, and economic feasibility.

5.7 Conclusions

The study successfully demonstrated that recycled PET material generated from rejected PET preforms can be incorporated into virgin PET resin for stretch blow moulding applications.

The results showed that increasing recycled PET content gradually reduced stiffness, thermal stability, and structural performance due to polymer degradation effects. However, recycled PET blends containing 10% and 15% recycled material maintained acceptable thermo-mechanical performance suitable for industrial processing conditions.

COMSOL Multiphysics simulations successfully evaluated the behavior of the recycled PET blends during reheating and bottle inflation stages. The simulation results showed that the 15% recycled PET blend achieved the best balance between structural performance, sustainability improvement, and economic feasibility.

The study therefore demonstrated the feasibility of implementing recycled PET integration strategies within industrial PET bottle manufacturing systems to improve sustainability and reduce manufacturing waste.

CHAPTER 6: CONCLUSIONS AND RECOMMENDATIONS

6.1 Conclusions

This research project investigated the thermodynamic and mechanical implications of integrating varying percentages of recycled PET (10%, 15%, and 20%) into 2-liter bottle preforms. By analyzing the direct correlation between processing temperature profiles and locked-in residual Von Mises stresses, several critical conclusions can be drawn regarding material behavior, structural integrity, and manufacturing feasibility.

First, the primary body of the preform (0 to 80 mm from the base) exhibits highly predictable and stable behavior across all configurations. Maintaining a uniform heating zone of 115 degrees Celsius ensures that the polymer matrix is adequately conditioned above its glass transition temperature, allowing for uniform material stretch. However, the introduction of higher rPET content imposes an inherent mechanical penalty. Even under identical thermal conditions in this lower zone, the 20% rPET blend exhibits significantly higher base residual stresses (10 to 13 megapascals) compared to the 10% and 15% blends (0 to 7 megapascals). This confirms that increased recycled content fundamentally alters the polymer melt rheology, likely due to a combination of altered molecular weight distribution, structural micro-impurities, or residual crystalline nodes that resist uniform fluid flow.

Second, a severe spatial thermal shock zone exists precisely at the 90 to 100 mm height range, where the preform transitions from the intense heating zone to localized cooling. This abrupt spatial temperature drop-off causes rapid, uneven volumetric contraction between the outer skin and the inner core of the polymer matrix. As a direct consequence, a massive global stress spike is locked into the material at this coordinate across all blends, peaking at 15 megapascals for both the 10% and 15% blends, and reaching a critical maximum of 18 megapascals for the 20% blend.

Finally, the 20% rPET blend represents a severe risk to product dimensional stability in the upper preform segments. Industrial processing standards specify that the preform neck finish (120 to 130 mm) must not exceed a maximum temperature of 20 degrees Celsius during handling to prevent thread distortion and physical sagging. While the 10% and 15% blends cool successfully to this

threshold, the 20% rPET preform retains an elevated temperature of 40 degrees Celsius at the neck while simultaneously holding 2 megapascals of residual stress. Consequently, operating a 20% rPET blend under unmodified standard production parameters is structurally unviable and will lead to severe geometric defects, neck ovality, and sealing failures in the finished container.

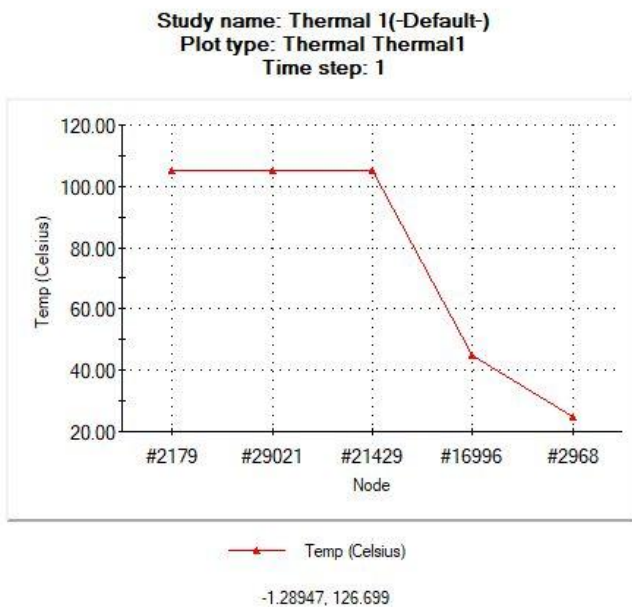
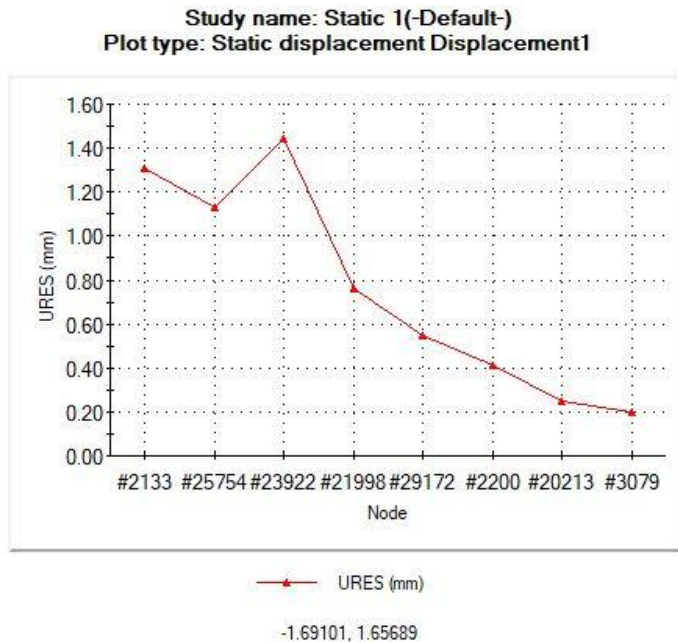
6.2 Recommendations

To successfully leverage high-percentage rPET blends without compromising the structural or aesthetic quality of the final manufactured packaging, the following engineering and process control interventions are recommended:

1. **Implement Localized High-Velocity Neck Cooling:** To counteract the severe heat retention observed in the 20% rPET blend at the 120 to 130 mm zone, the stretch blow molding machinery must be retrofitted with targeted cooling rings or high-velocity chilled air nozzles. Forcing the neck temperature down from 40 degrees Celsius to the mandatory maximum threshold of 20 degrees Celsius prior to the high-pressure blow sequence is essential to lock in the dimensions of the bottle finish and eliminate thread warpage.
2. **Optimize Heating Profiles to Smooth the Transition Zone Gradient:** To mitigate the severe 18 megapascals Von Mises stress spike occurring at the 90 to 100 mm height range, lamp configurations in the infrared heating oven must be adjusted. By establishing a more gradual, staggered thermal step-down rather than an abrupt drop-off, the spatial temperature gradient can be minimized. This will reduce differential volumetric contraction and prevent excessive residual shear stresses from locking into the transition zone.
3. **Adjust Stretching Speed and Pressure Set-Points for Higher Viscosity:** Because the 20% rPET matrix exhibits a higher inherent resistance to flow (as evidenced by its elevated base stress levels), the mechanical stretch rod speed and the pre-blow pressure timing should be dynamically tuned. Lowering the initial stretching speed slightly will allow the more rigid recycled polymer chains to orient smoothly without tearing or inducing excessive localized strain hardening.
4. **Conduct Further Rheological and Crystallization Testing:** It is strongly recommended to expand this research by conducting Differential Scanning Calorimetry (DSC) and Melt Flow Index (MFI) testing on the rPET blends. Characterizing the exact crystallization kinetics and viscosity variances introduced by the 10%, 15%, and 20% recycled batches will provide deeper data to refine future finite element analysis (FEA) models and optimize processing windows.

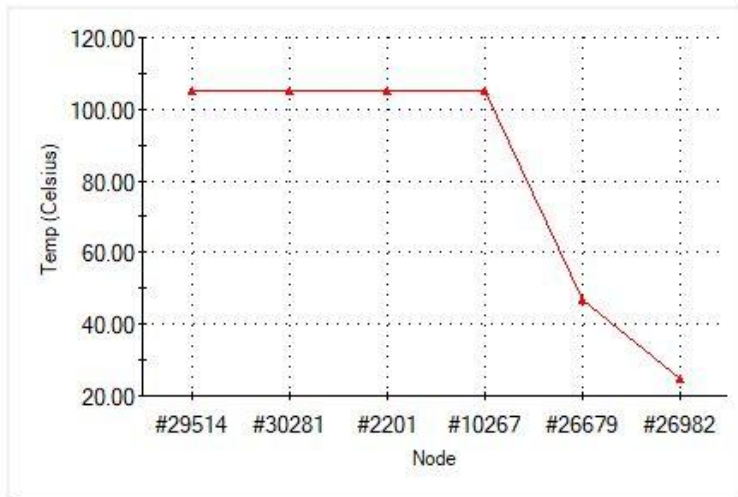
Appendix : Screenshot from SolidWorks and Comsol

Graphs For 15% rPET



Graphs For 15% rPET

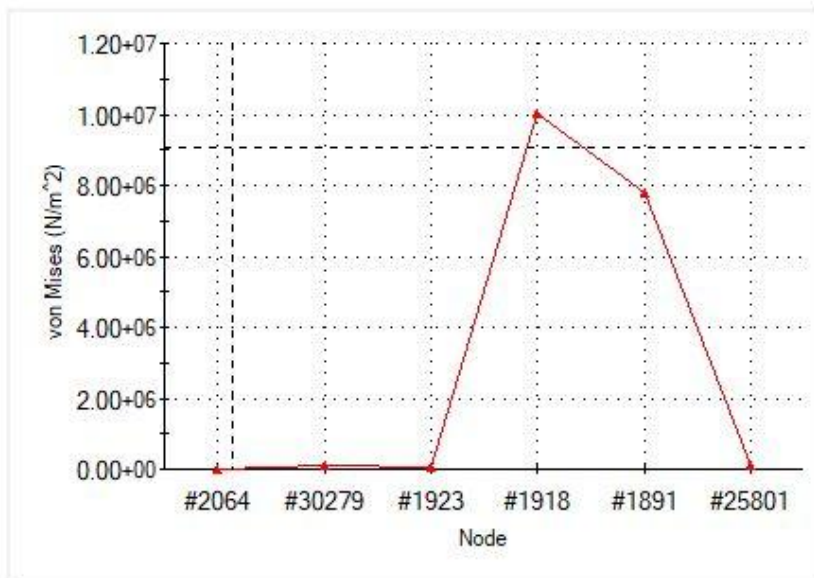
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Time step: 1



Temp (Celsius)

0, 0

Study name: Static 1(-Default-)
Plot type: Static nodal stress Stress1

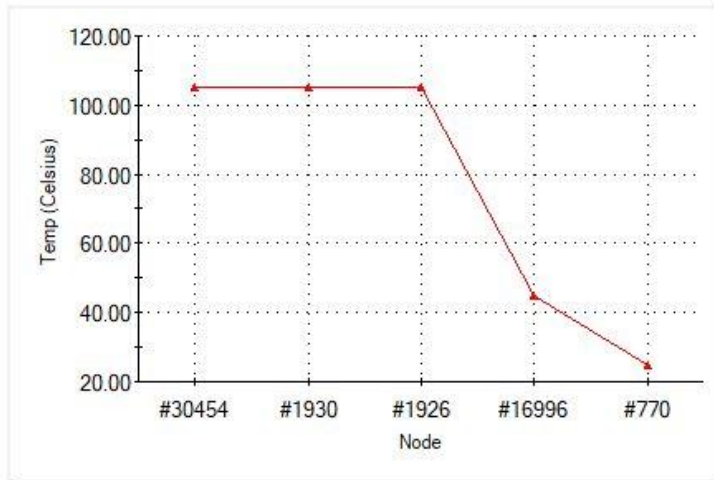


von Mises (N/m^2)

0.139053, 9.06667e+06

Graphs For 10% rPET

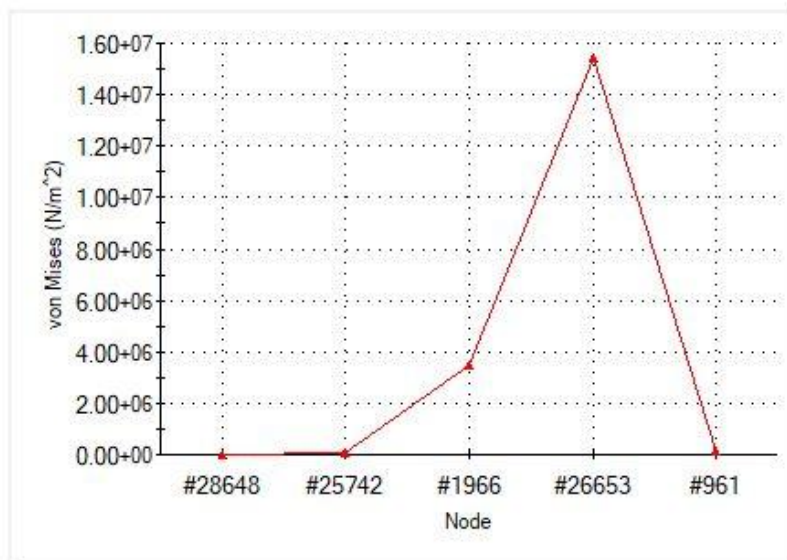
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Plot type: Thermal Thermal1
Time step: 1



Temp (Celsius)

0.611111, 127.656

Study name: Static 1(-Default-)
Plot type: Static nodal stress Stress1



von Mises (N/m^2)

0.121302, 1.45067e+07

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